

Class XI: Fundamentals of Physical Geography.

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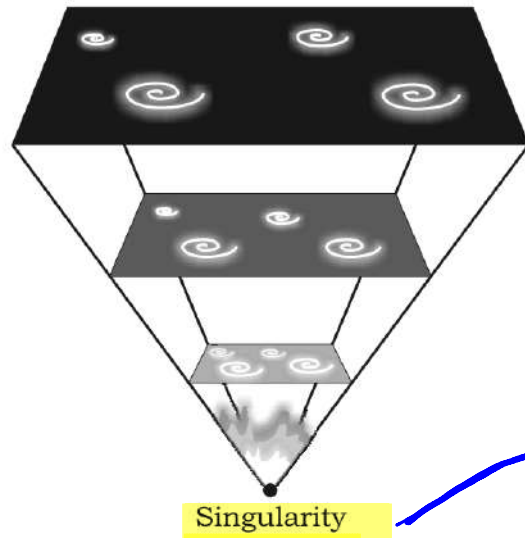


Figure 2.1 : The Big Bang

→ temp;
→ density
Expanding Universe
By Lemaitre

Geological Time Scale

Eons	Era	Period	Epoch	Age / Years Before Present	Life / Major Events
	Cainozoic (From 65 million years to the present times)	Quaternary	Holocene	0 - 10,000	Modern Man
			Pleistocene	10,000 - 2 million	Homo Sapiens
		Tertiary	Pliocene	2 - 5 million	Early Human Ancestor
			Miocene	5 - 24 million	Ape: Flowering Plants and Trees
			Oligocene	24 - 37 Ma	Anthropoid Ape
	Palaeozoic 245 - 570 Million	Carboniferous	Eocene	37 - 58 Million	Rabbits and Hare
			Palaeocene	57 - 65 Million	Small Mammals : Rats - Mice
	Mesozoic 65 - 245 Million Mammals	Cretaceous		65 - 144 Million	Extinction of Dinosaurs
				144 - 208 Million	Age of Dinosaurs
		Carboniferous		208 - 245 Million	Frogs and turtles
				245 - 286 Million	Reptile dominate-replace amphibians
				286 - 360 Million	First Reptiles: Vertebrates: Coal beds
				360 - 408 Million	Amphibians
				408 - 438 Million	First trace of life on land: Plants
				438 - 505 Million	First Fish
	Proterozoic Archean	Pre-Cambrian		505 - 570 Million	No terrestrial Life : Marine Invertebrate
	Hadean			570 - 2,500 Million	Soft-bodied arthropods
				2,500 - 3,800 Million	Blue green Algae: Unicellular bacteria
				3,800 - 4,800 Million	Oceans and Continents form - Ocean and Atmosphere are rich in Carbon dioxide
	Origin of Stars			5,000 Million	Origin of the sun
				12,000 Million	Origin of the universe
				13,700 Million	

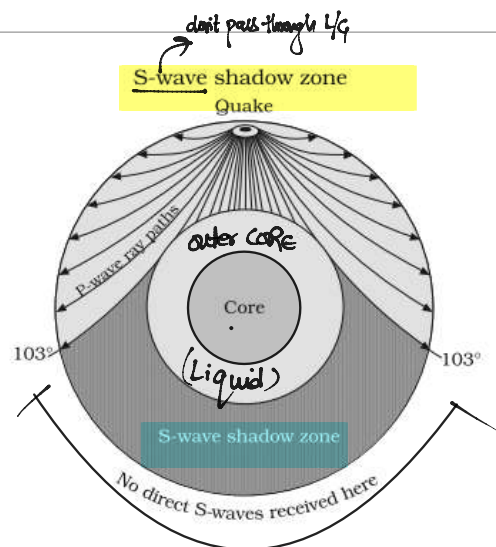
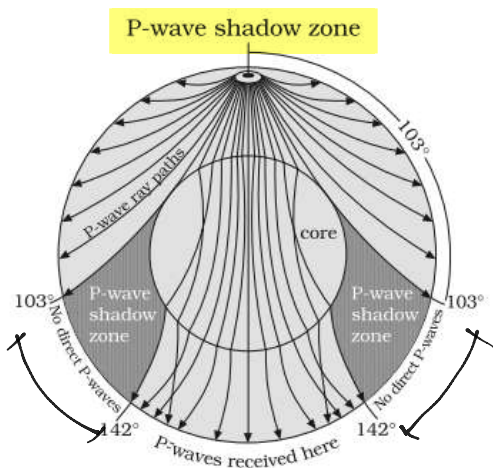
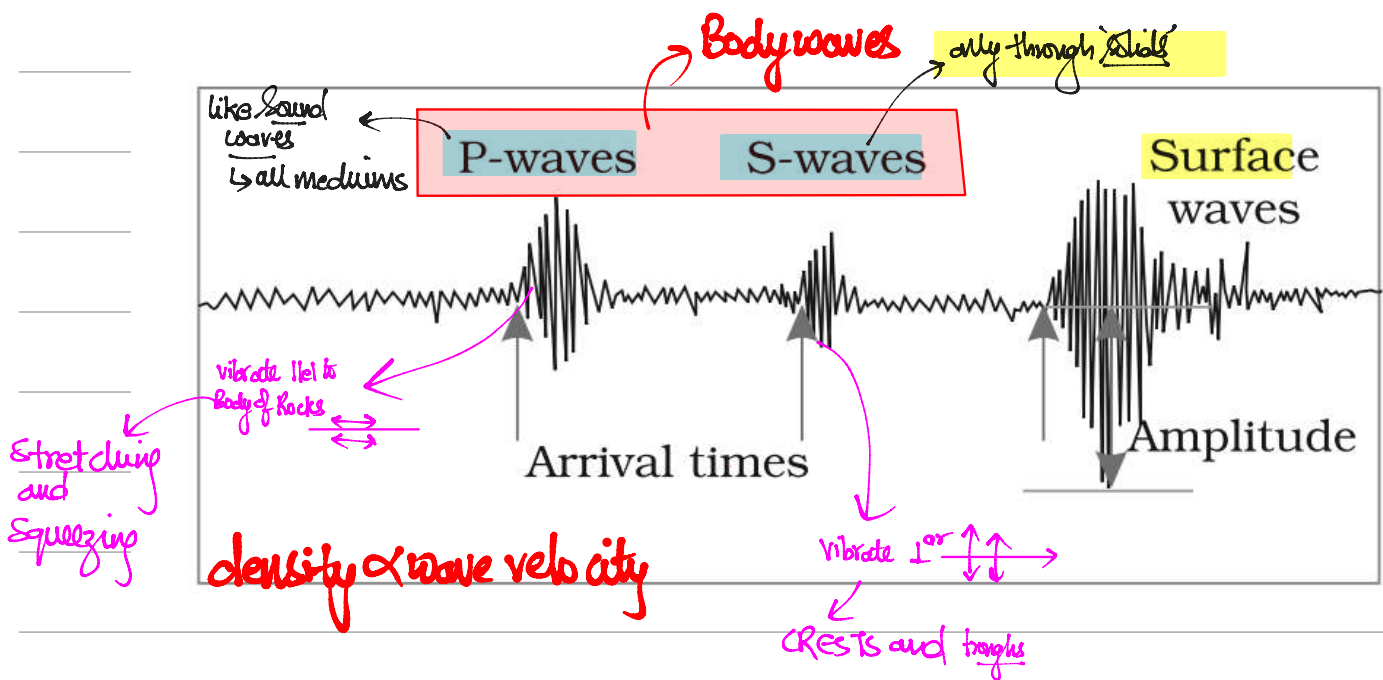
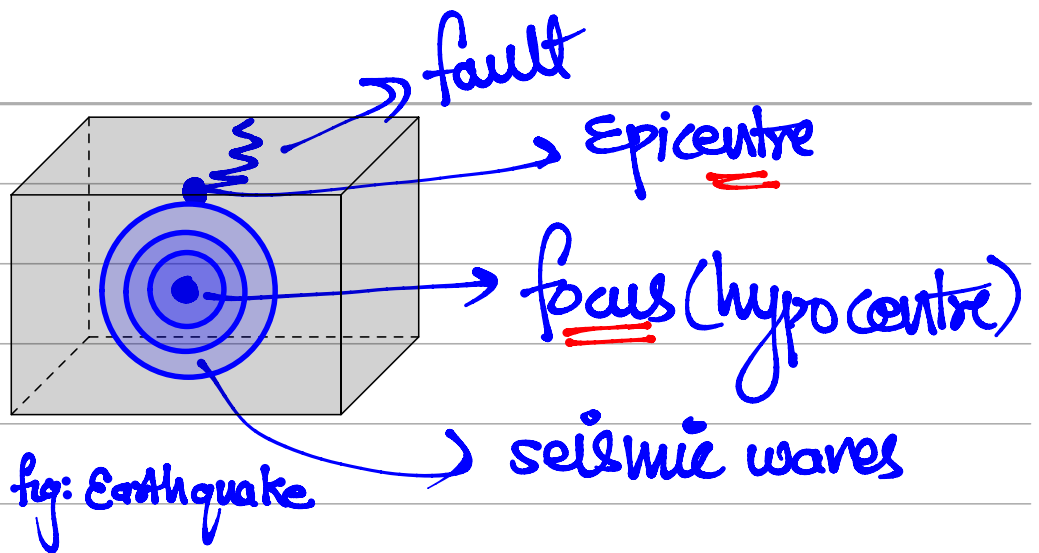


Figure 3.2 (a) and (b) : Earthquake Shadow Zones

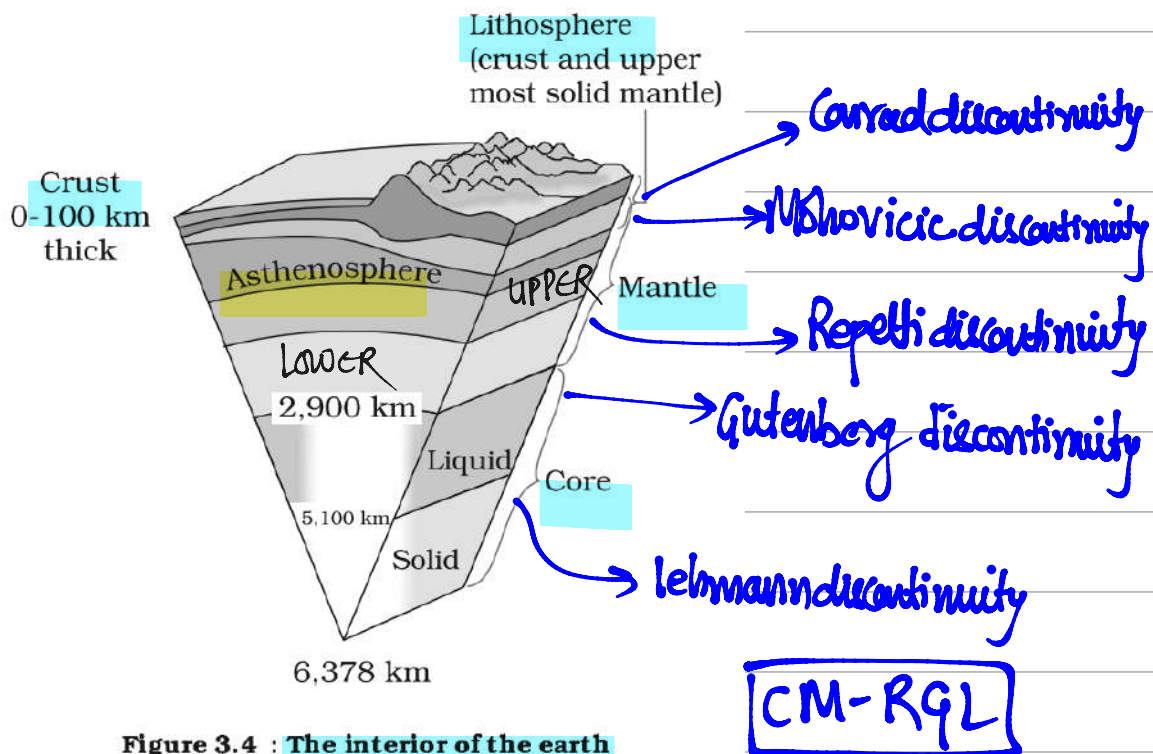


Figure 3.4 : The interior of the earth

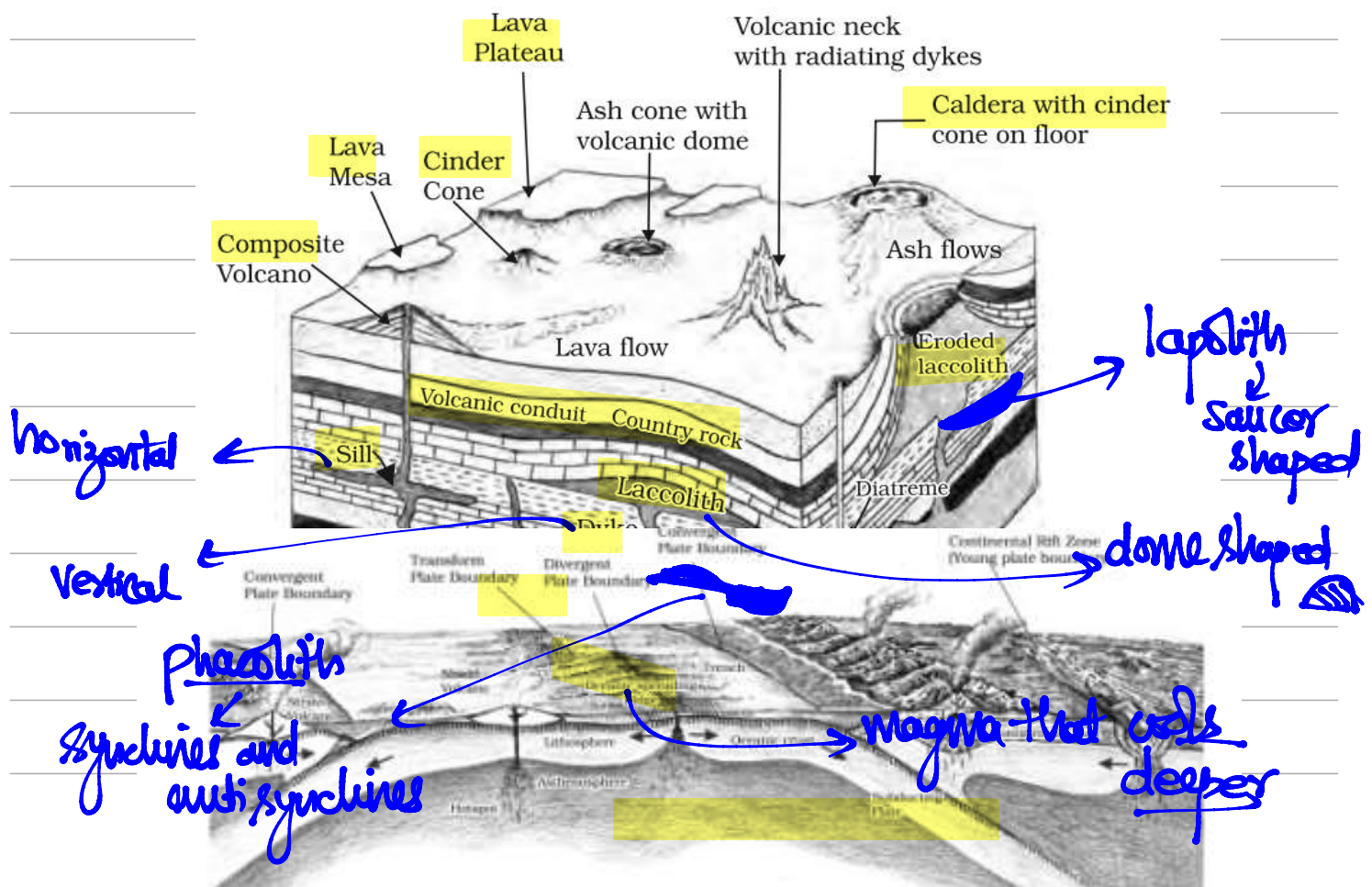
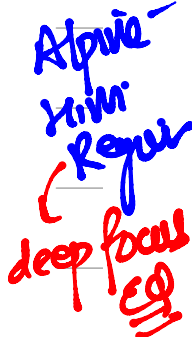


Figure 4.3 : Sea floor spreading



Figure 4.1 : Ocean Floor



Alpine Himalayan Region

Shallow focus

→ divergent plates

convergent
plate
boundary

subduction/slab pull



↳ HARRY HESS (1962)

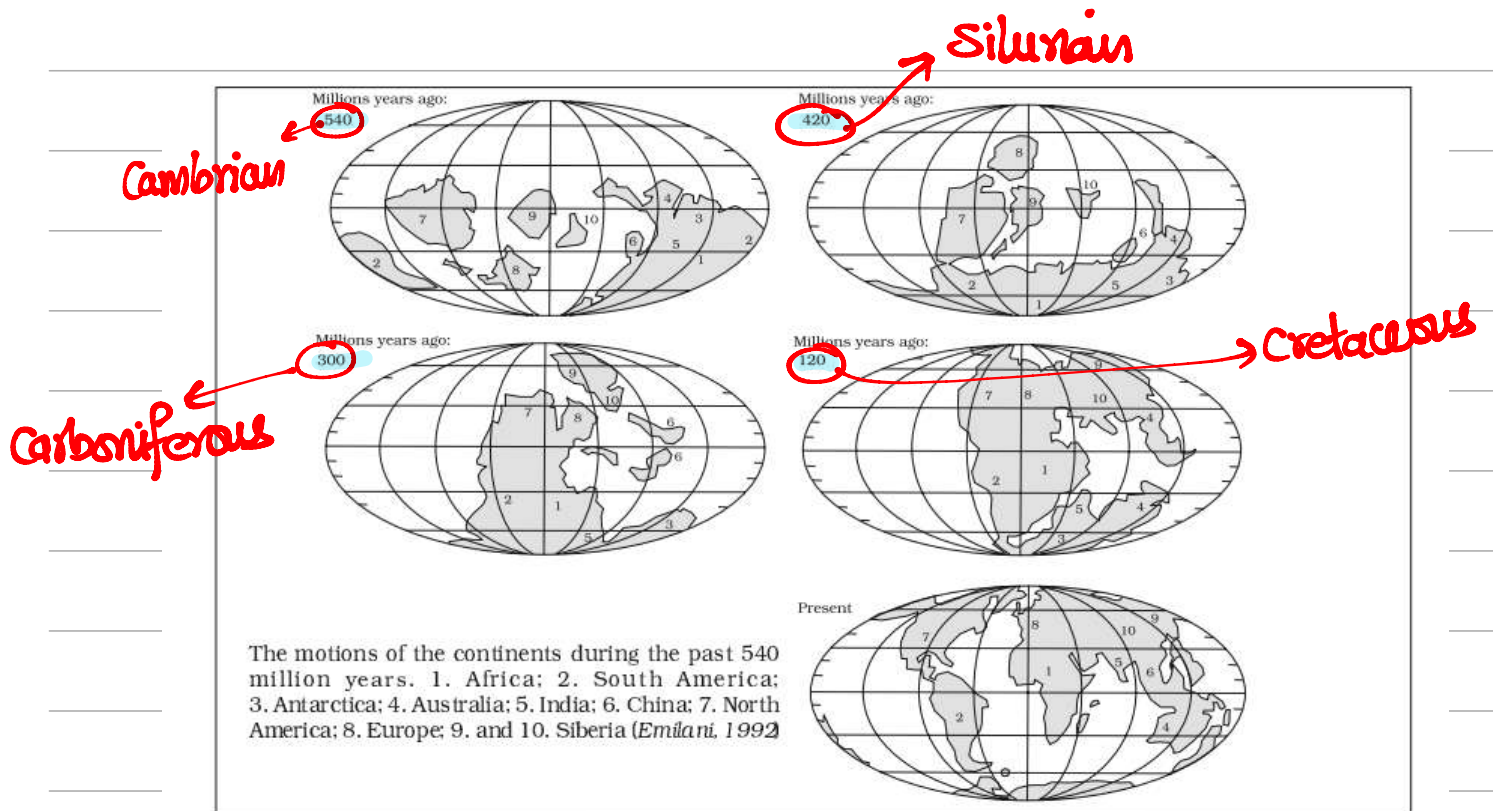


Figure 4.4 : Position of continents through geological past

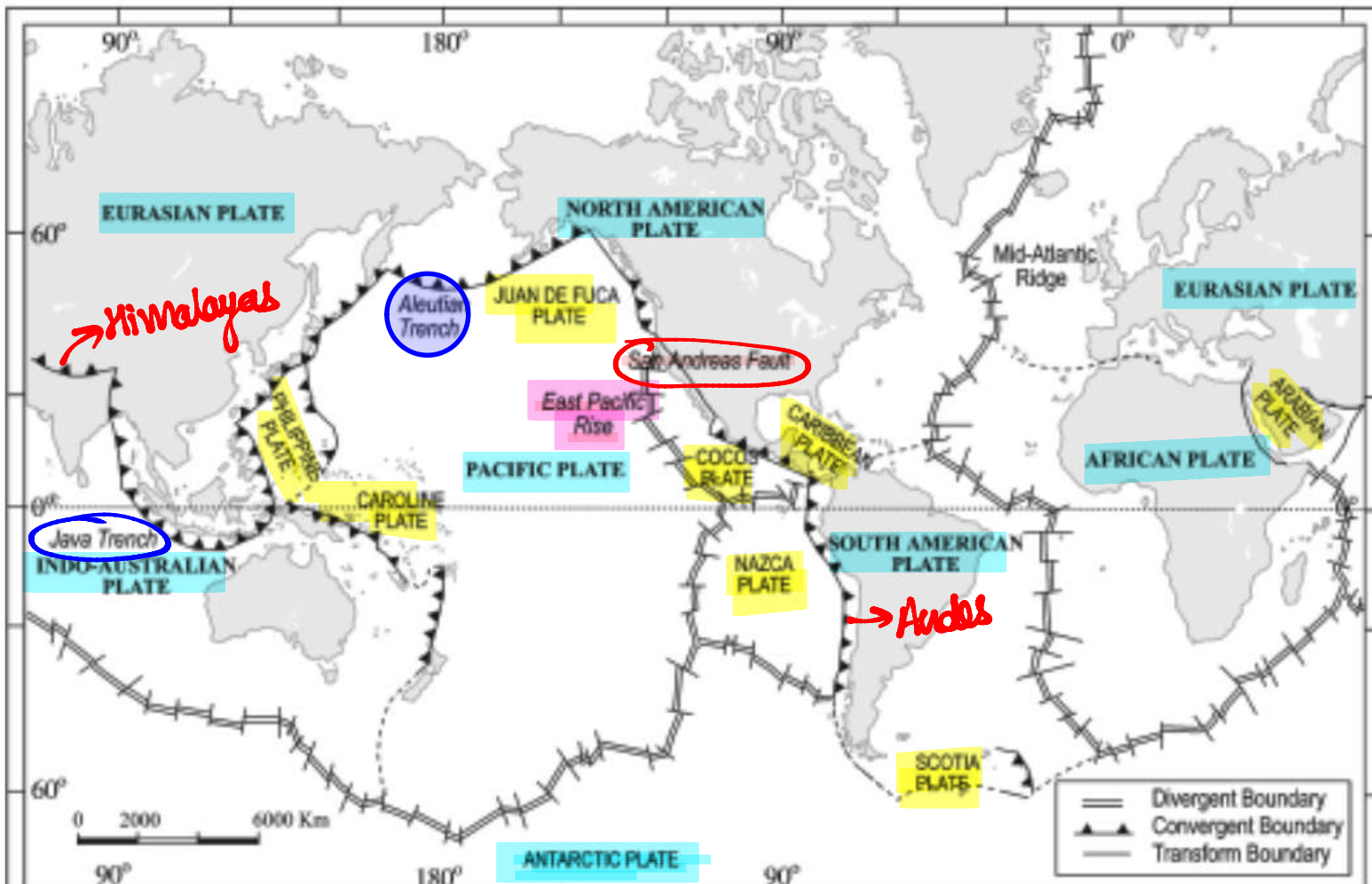


Figure 4.5 : Major and minor plates of the world

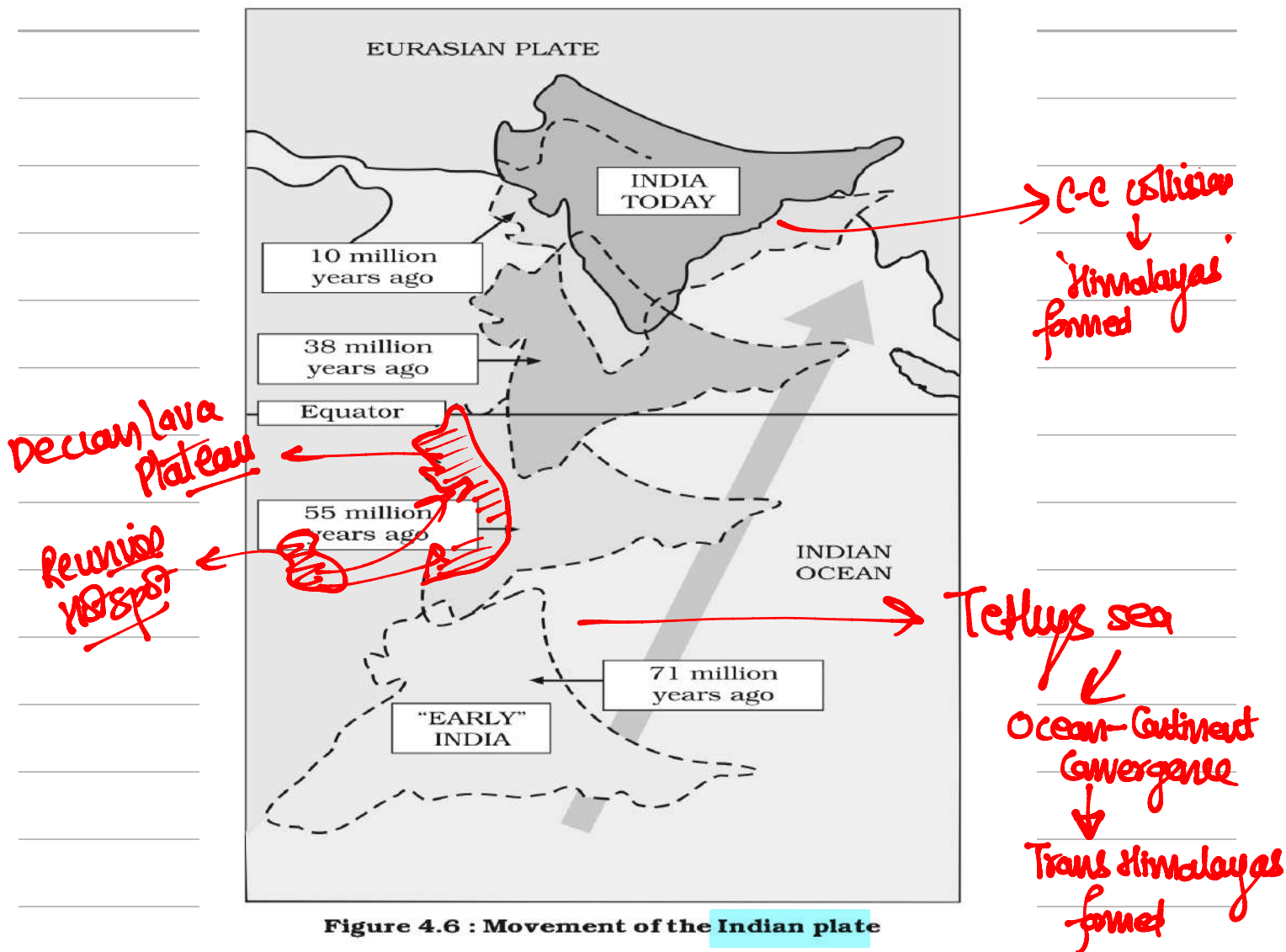


Figure 4.6 : Movement of the Indian plate

Table 5.1 : The Major Elements of the Earth's Crust

Sl. No.	Elements	By Weight(%)
1.	Oxygen	46.60
2.	Silicon	27.72
3.	Aluminium	8.13
4.	Iron	5.00
5.	Calcium	3.63
6.	Sodium	2.83
7.	Potassium	2.59
8.	Magnesium	2.09
9.	Others	1.41

O-SAI-
CSP-Mem
पास कर दो।

minerals

Half of Earth's crust is composed of Feldspar

Pyroxene forms 10% of Earth's crust

Si and O₂ are common element in feldspar

Mg
Ca
Al
Si
Fe

Amphibole - (7%) of Earth's crust
 Mica - (4%)

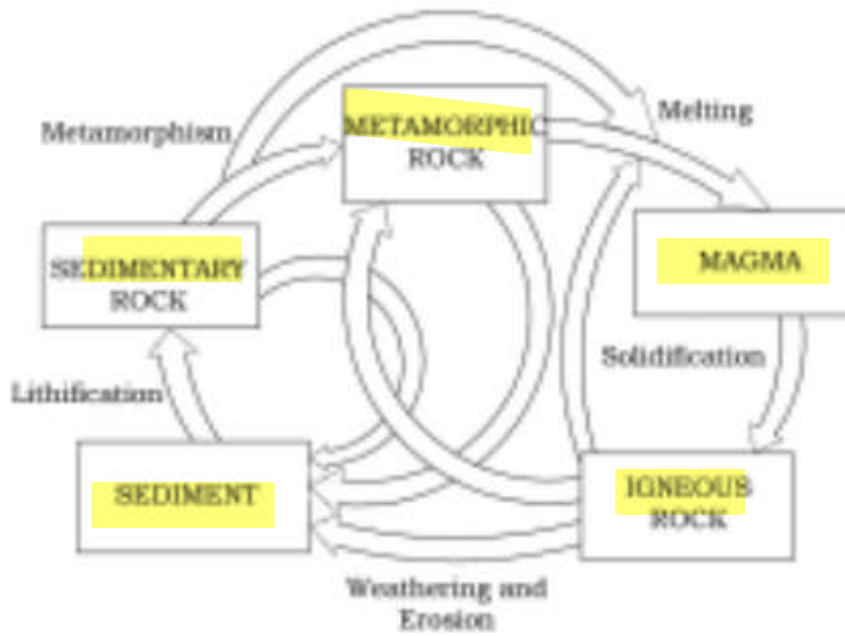


Fig 5.1 : Rock Cycle

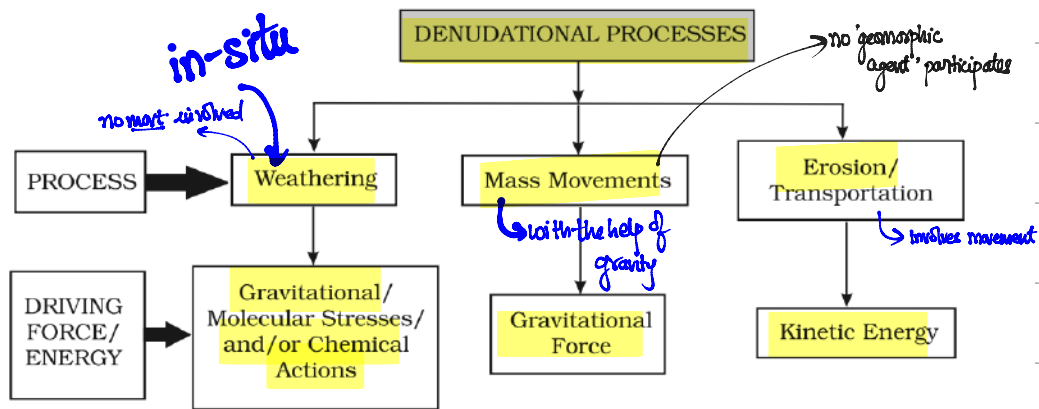
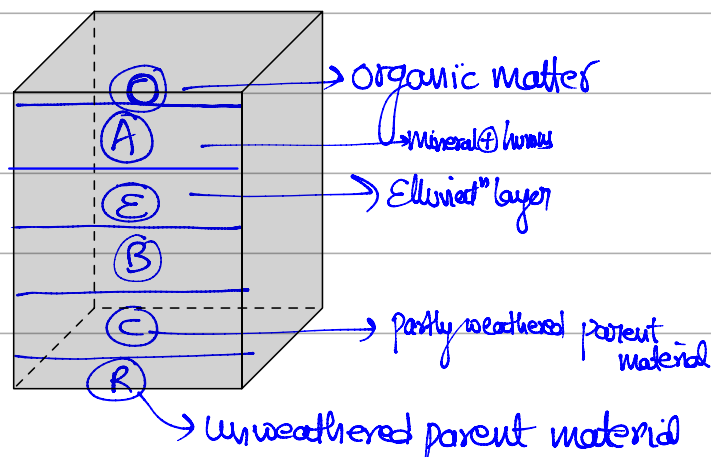


Figure 6.1 : Denudational processes and their driving forces



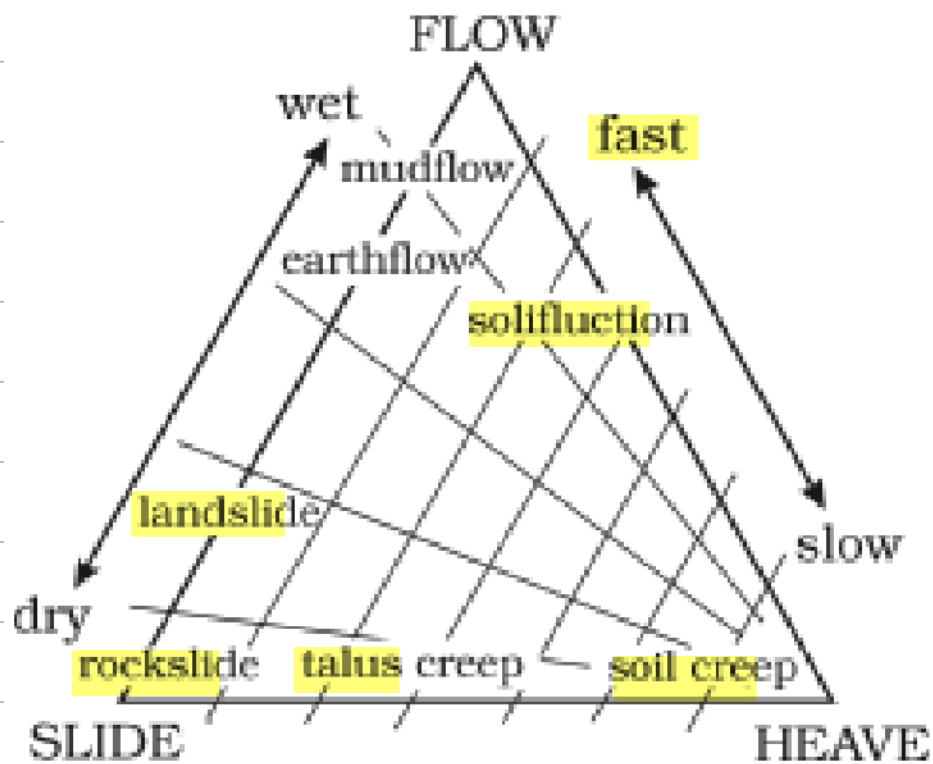


Figure 6.5 : Relationships among different types of mass movements, their relative rates of movement and moisture limits (after Whitehead, 2001)



Figure 7.3 : Paired and unpaired river terraces

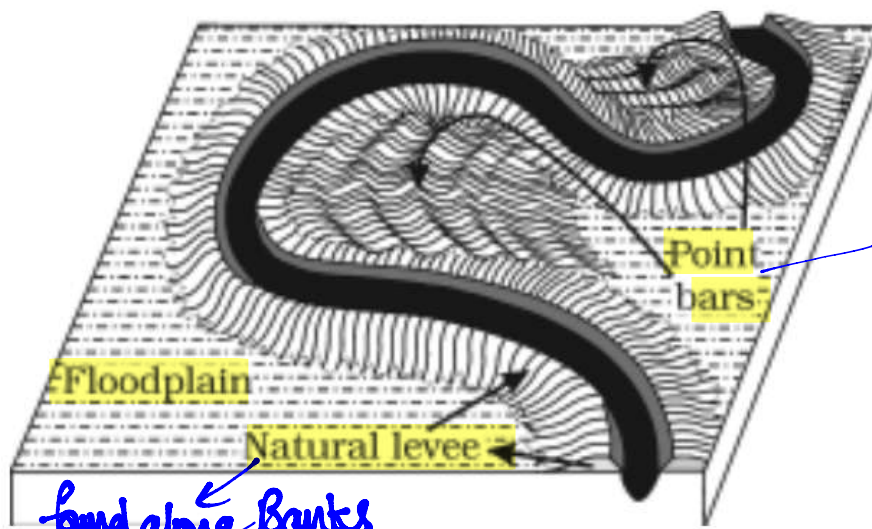


Figure 7.6 : Natural levee and point bars

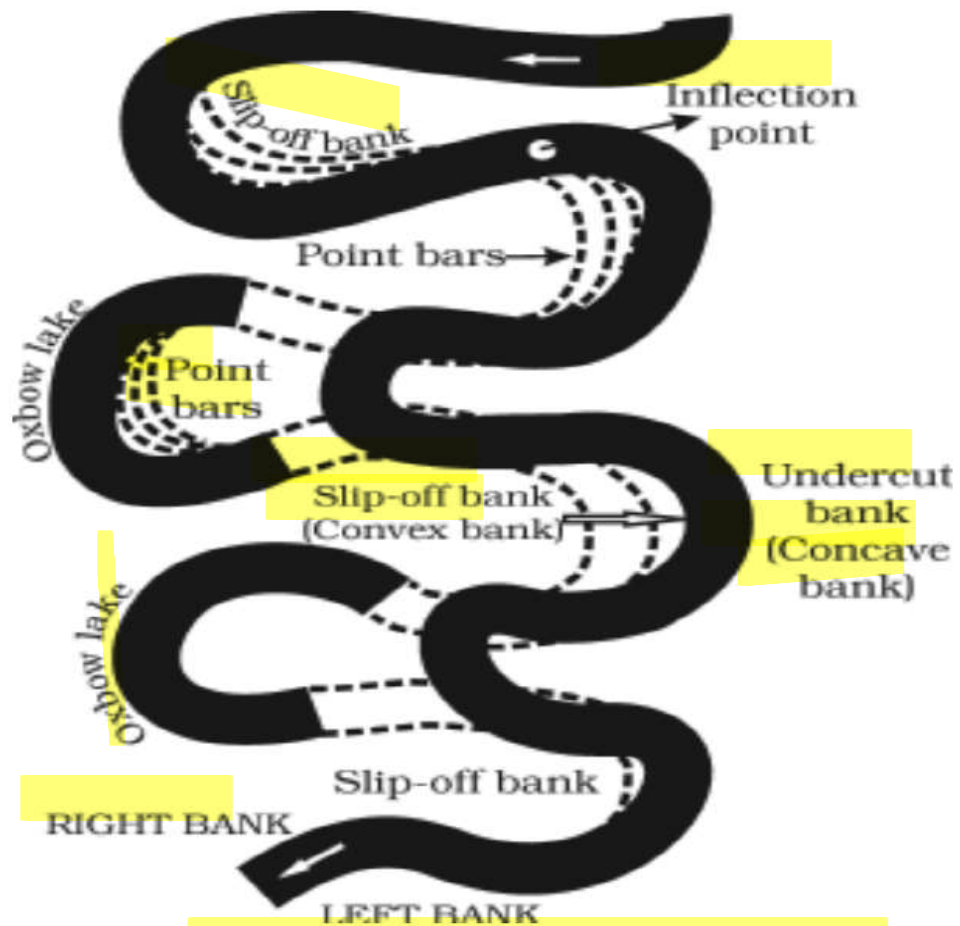


Figure 7.8 : Meander growth and cut-off loops and slip-off and undercut banks

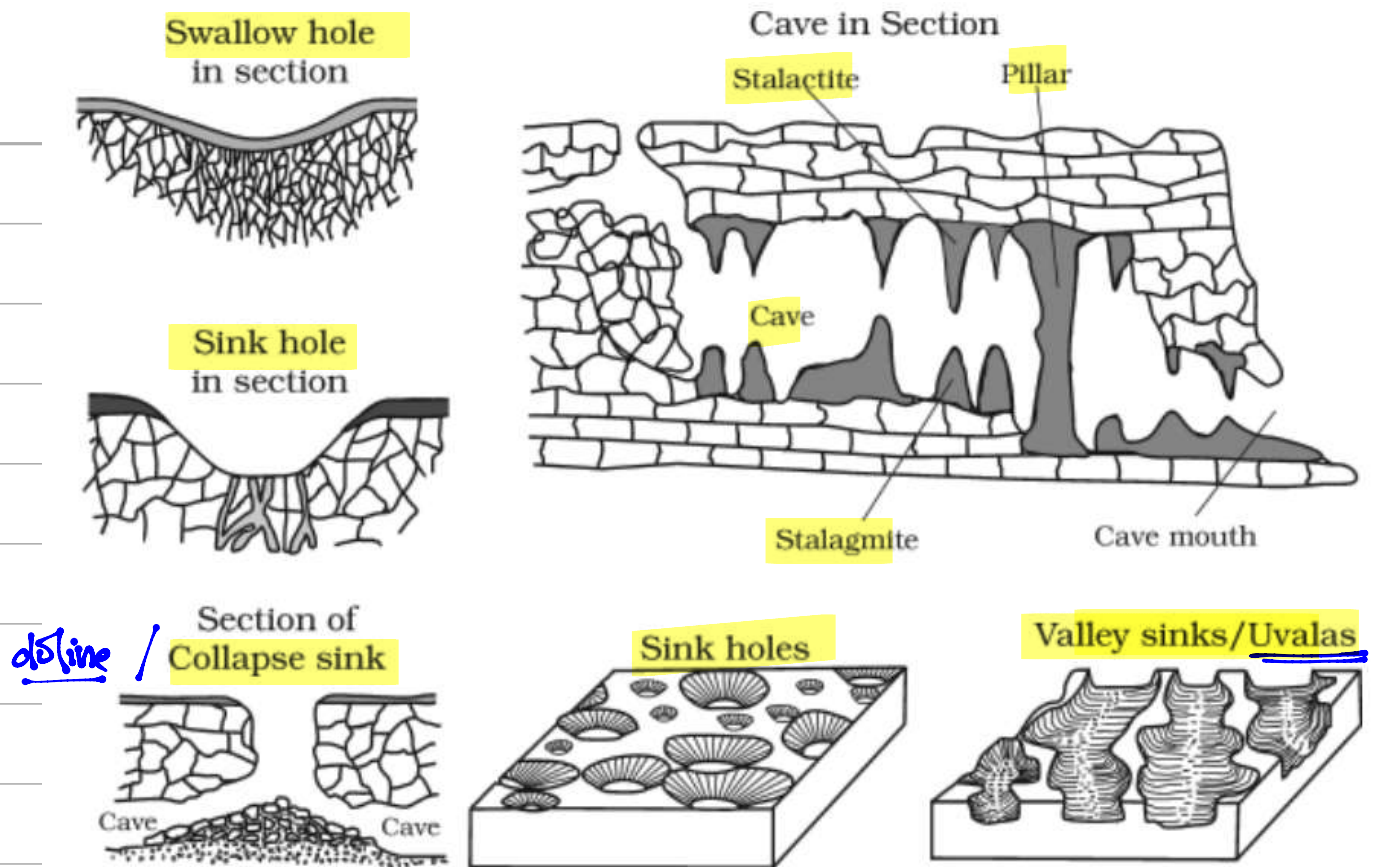


Figure 7.10: Various karst features

Glacial depositional landforms

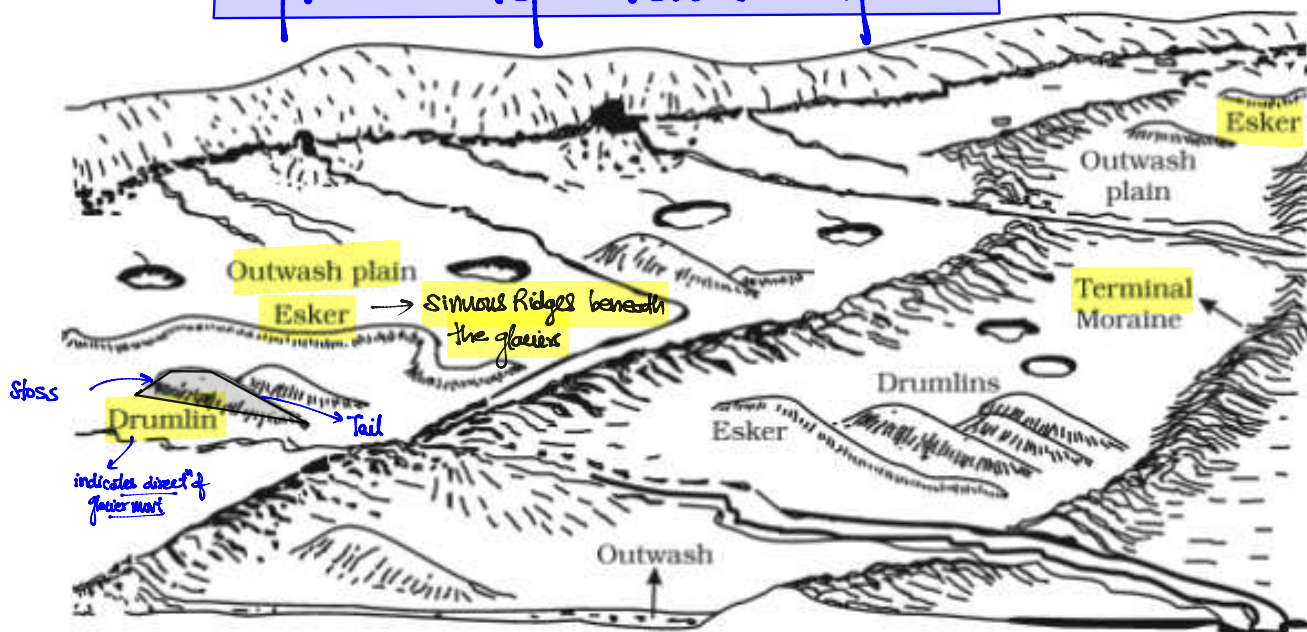


Figure 7.14: A panoramic diagram of glacial landscape with various depositional landforms (adapted and modified from Spencer, 1962)

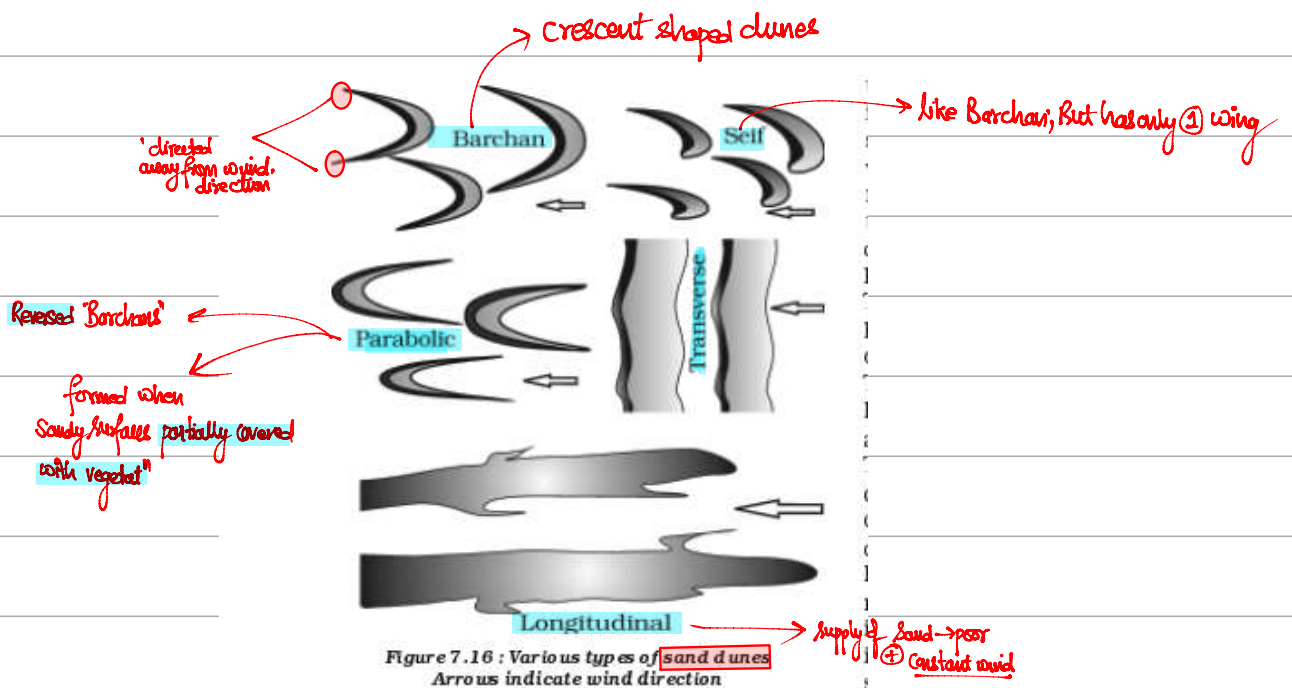
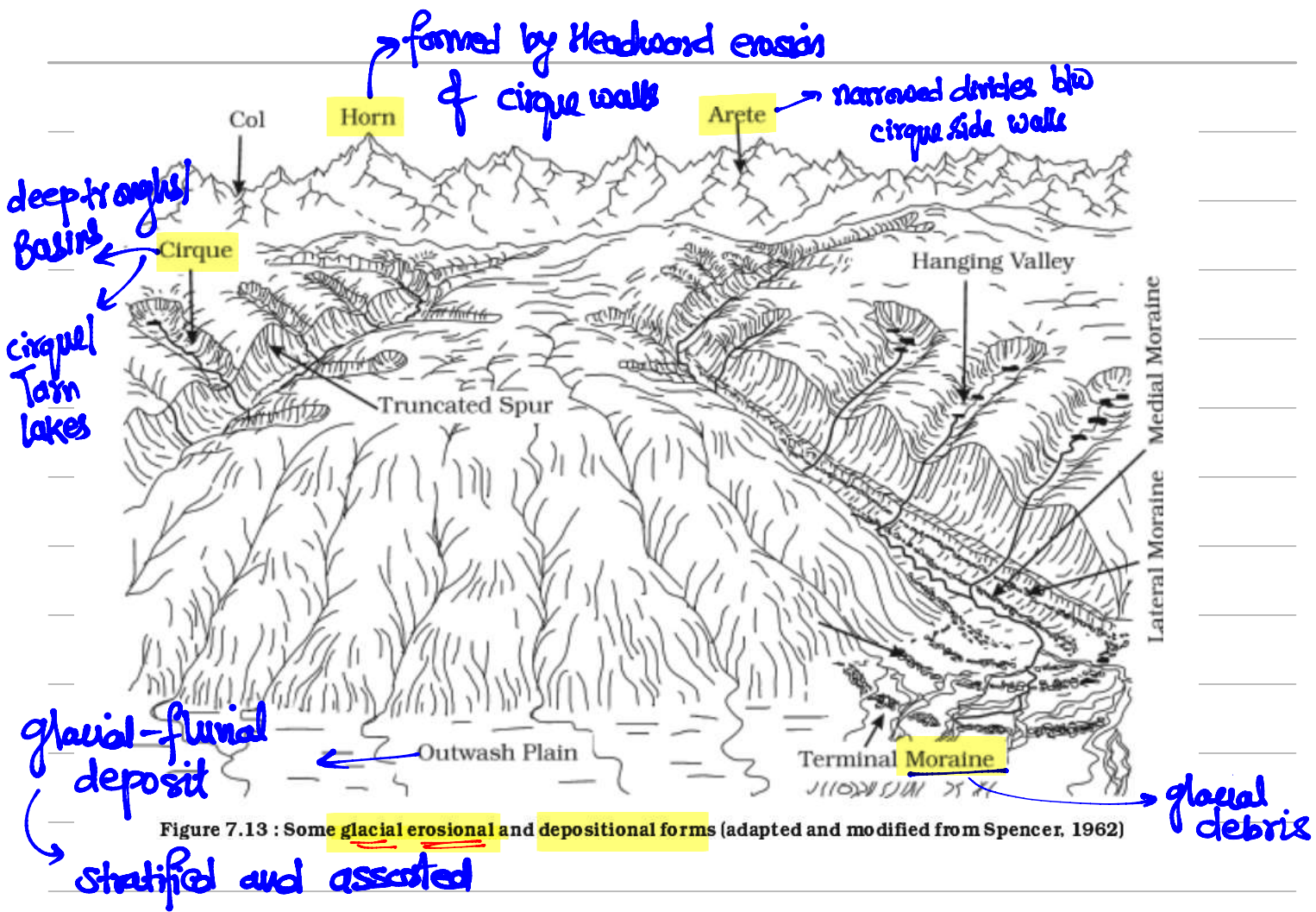


Table 8.1 : Permanent Gases of the Atmosphere

Constituent	Formula	Percentage by Volume
Nitrogen	N ₂	78.08
Oxygen	O ₂	20.95
Argon	Ar	0.93
Carbon dioxide	CO ₂	0.036
Neon	Ne	0.002
Helium	He	0.0005
Krypto	Kr	0.001
Xenon	Xe	0.00009
Hydrogen	H ₂	0.00005

Gases → NO AC in NH to krypto
 N₂ O₂ Argon CO₂ Neon He Xenon Gen-2 Hydrogen

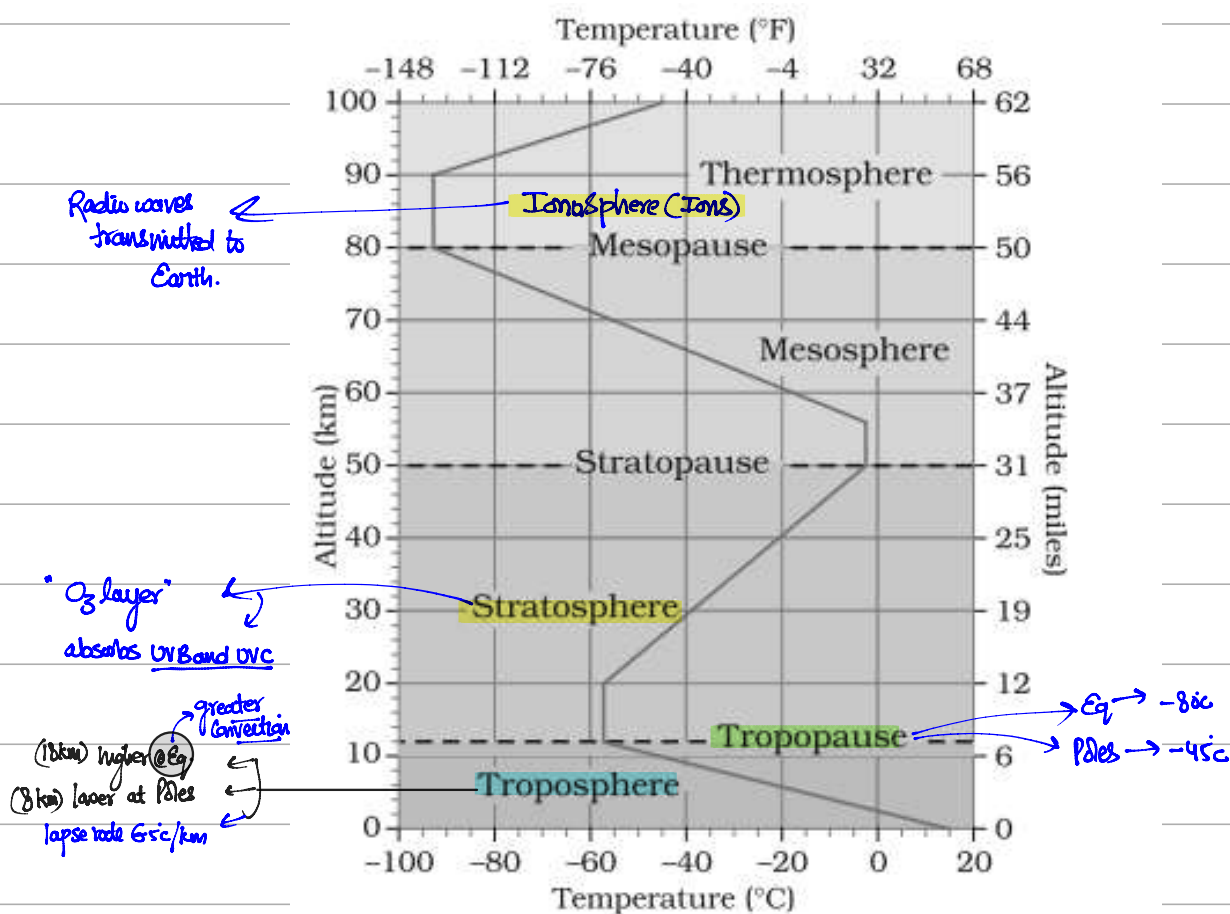


Figure 8.1 : Structure of atmosphere

Table 9.1 : Length of the Day in Hours and Minutes on Winter and Summer Solstices in the Northern Hemisphere

Latitude	0°	20°	40°	60°	90°
December 22 <i>winter Solstice</i>	12h 00m	10h 48m	9h 8m	5h 33m	0
June 21 <i>Summer Solstice</i>	12 h	13h 12m	14h 52m	18h 27m	6 months

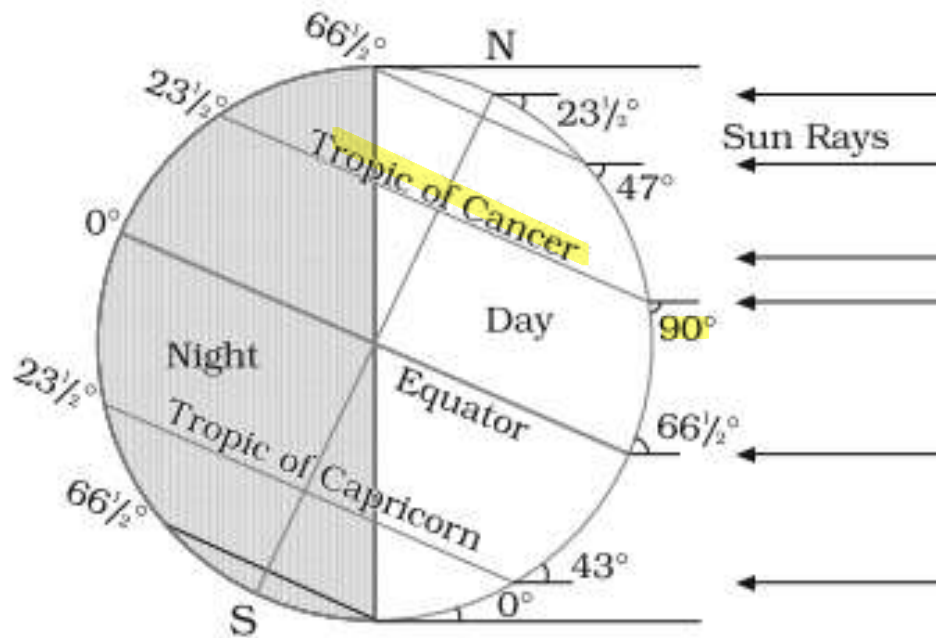


Figure 9.1 : Summer Solstice (21st June)

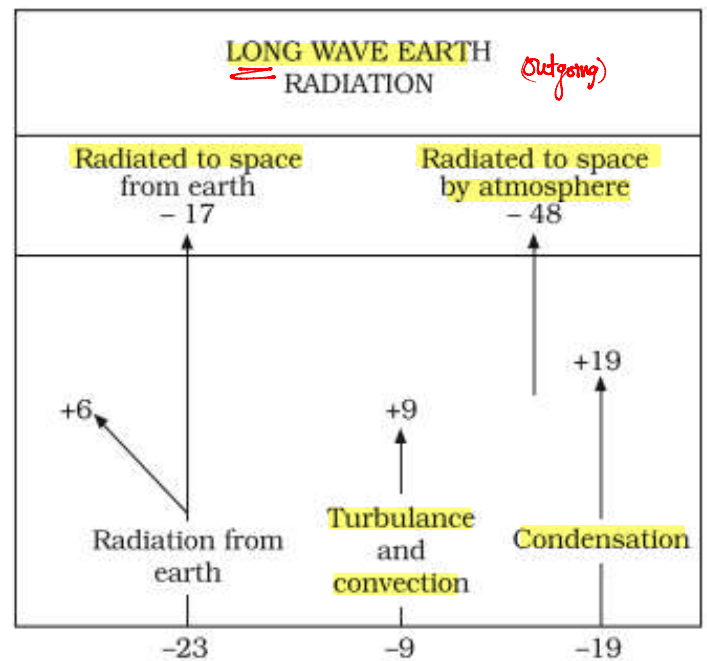
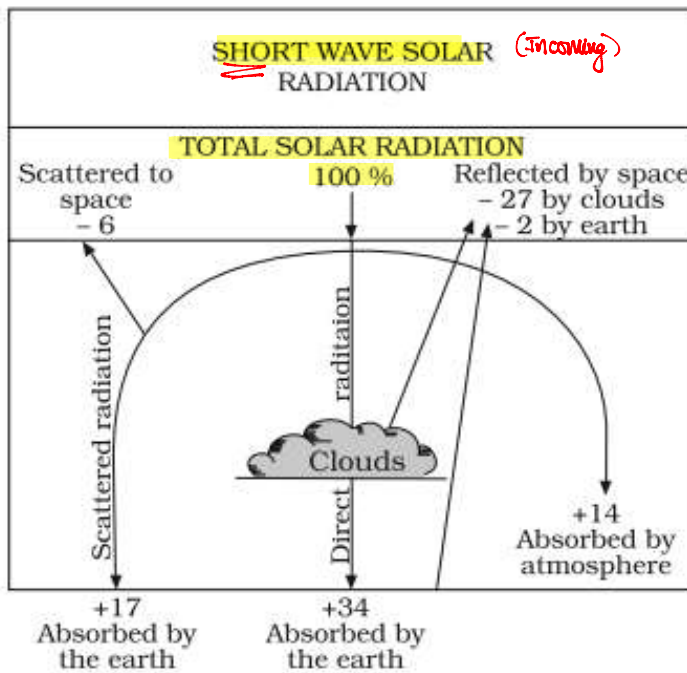
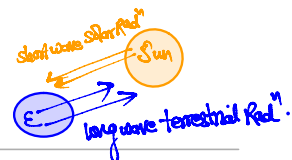


Figure 9.2 : Heat budget of the earth

↓
incoming short wave Solar Radiatⁿ = outgoing long wave earth Radiatⁿ



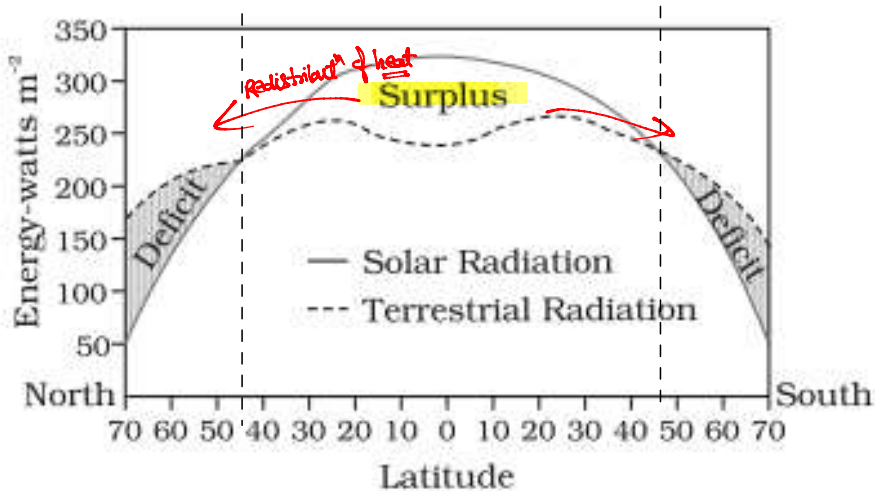


Figure 9.3 : Latitudinal variation in net radiation balance

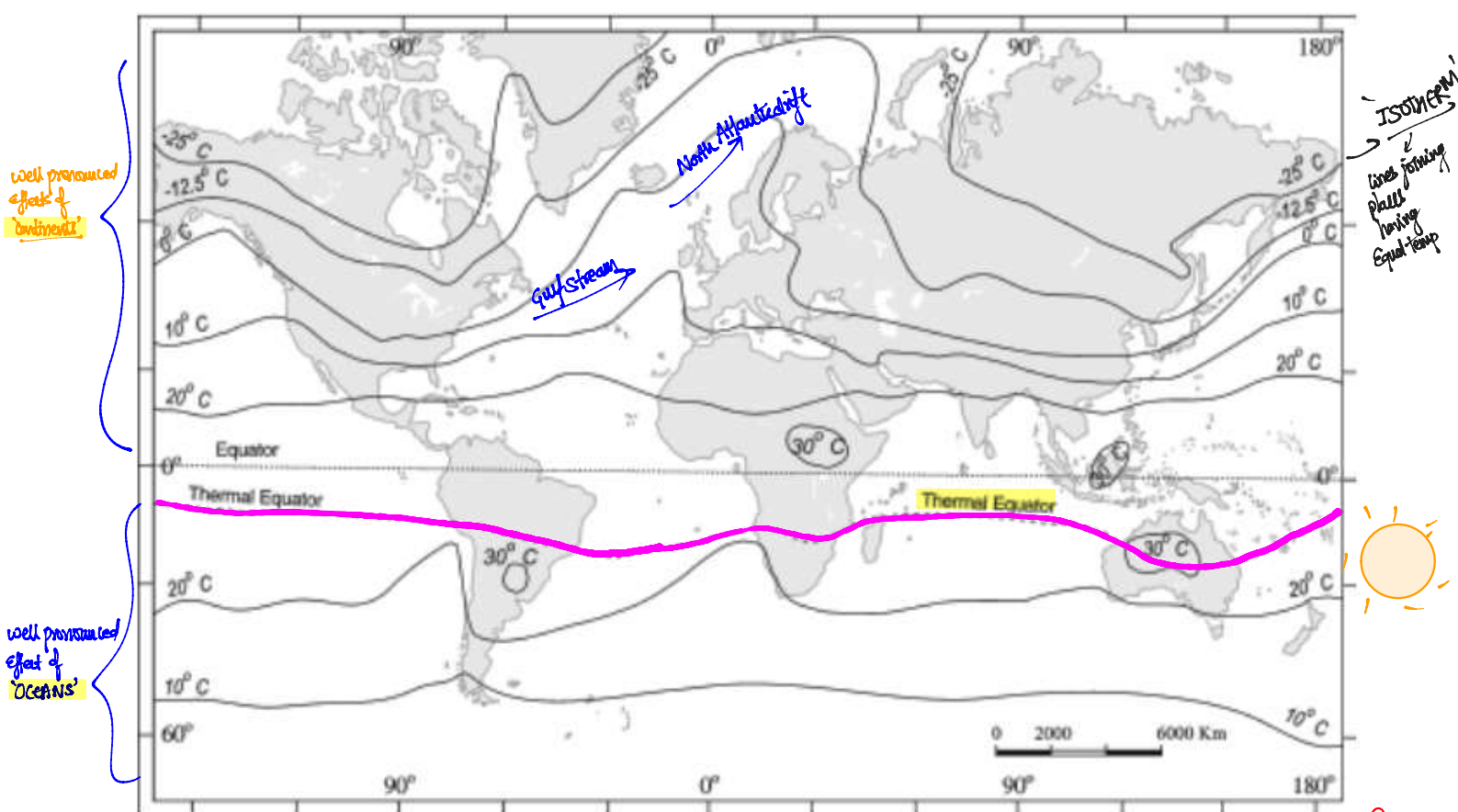


Figure 9.4 (a) : The distribution of surface air temperature in the month of January

→ greater deviation from Normal (that too more in N. Hemisphere)

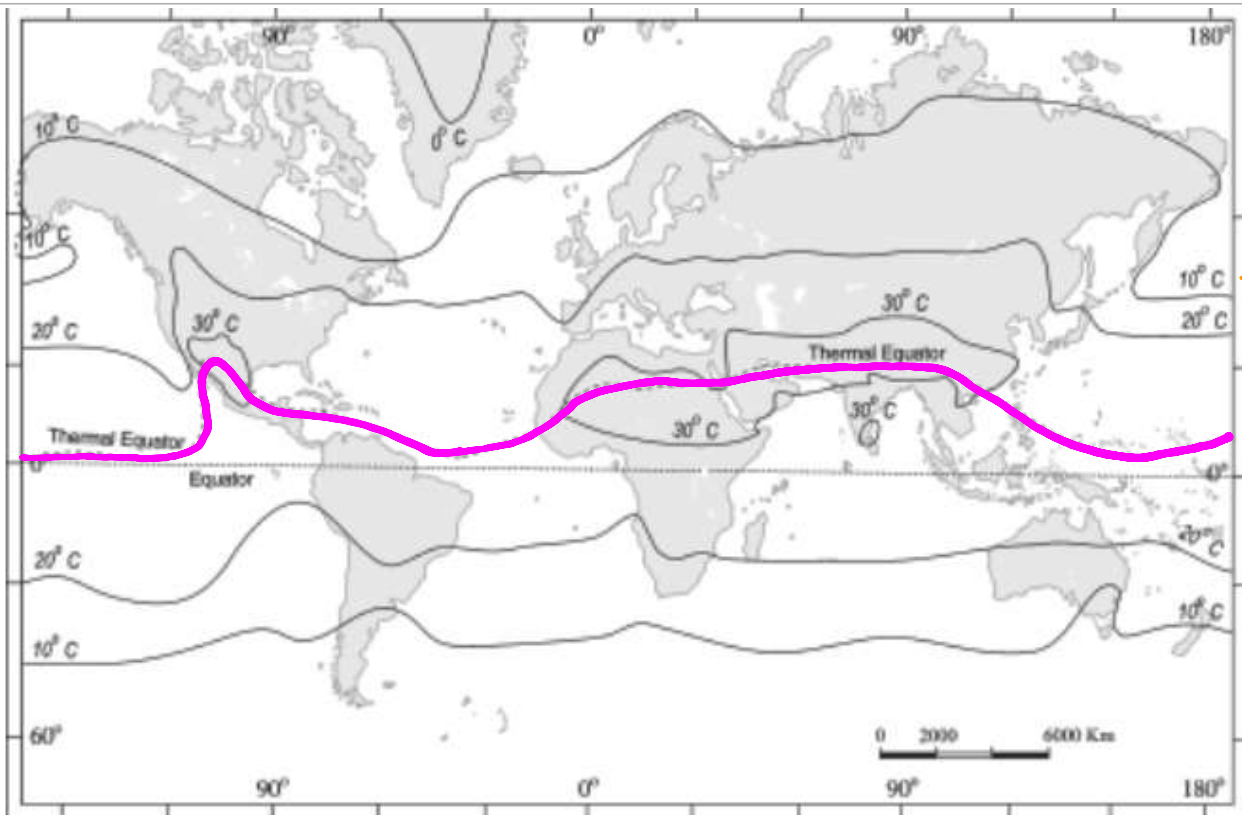


Figure 9.4 (b) : The distribution of surface air temperature in the month of July

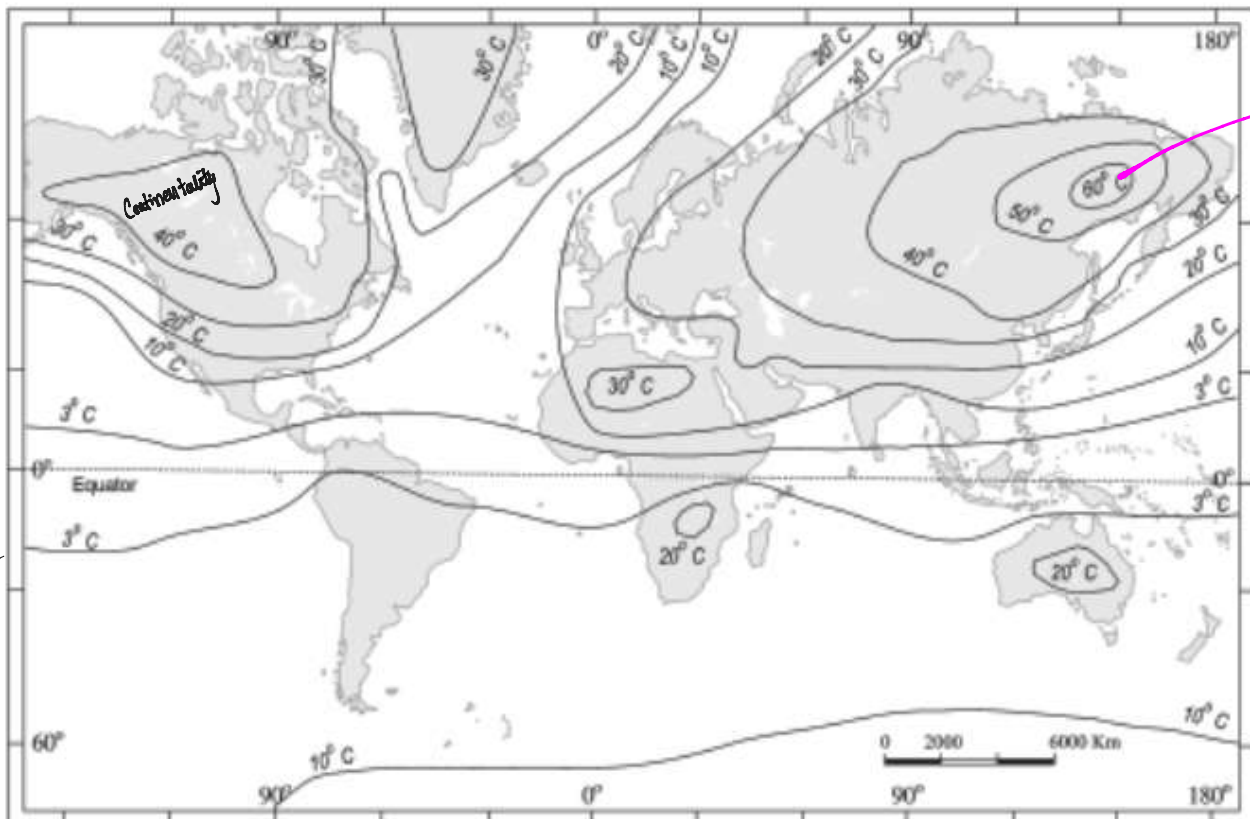


Figure 9.5 : The range of temperature between January and July

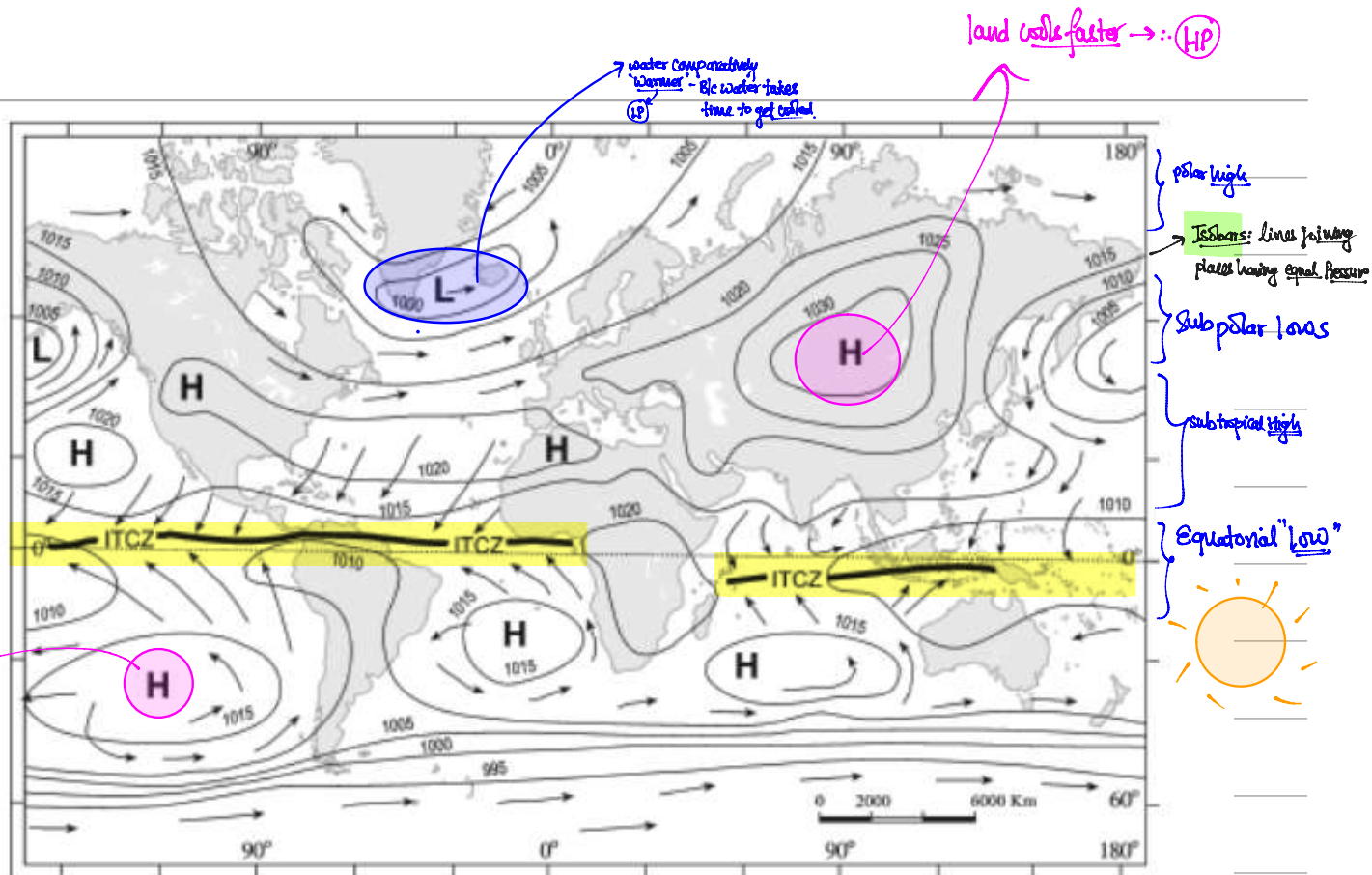


Figure 10.2 : Distribution of pressure (in millibars) — January

Winters @ N. Hemisphere
 over land → HP

∴ H₂O takes time to get heated;

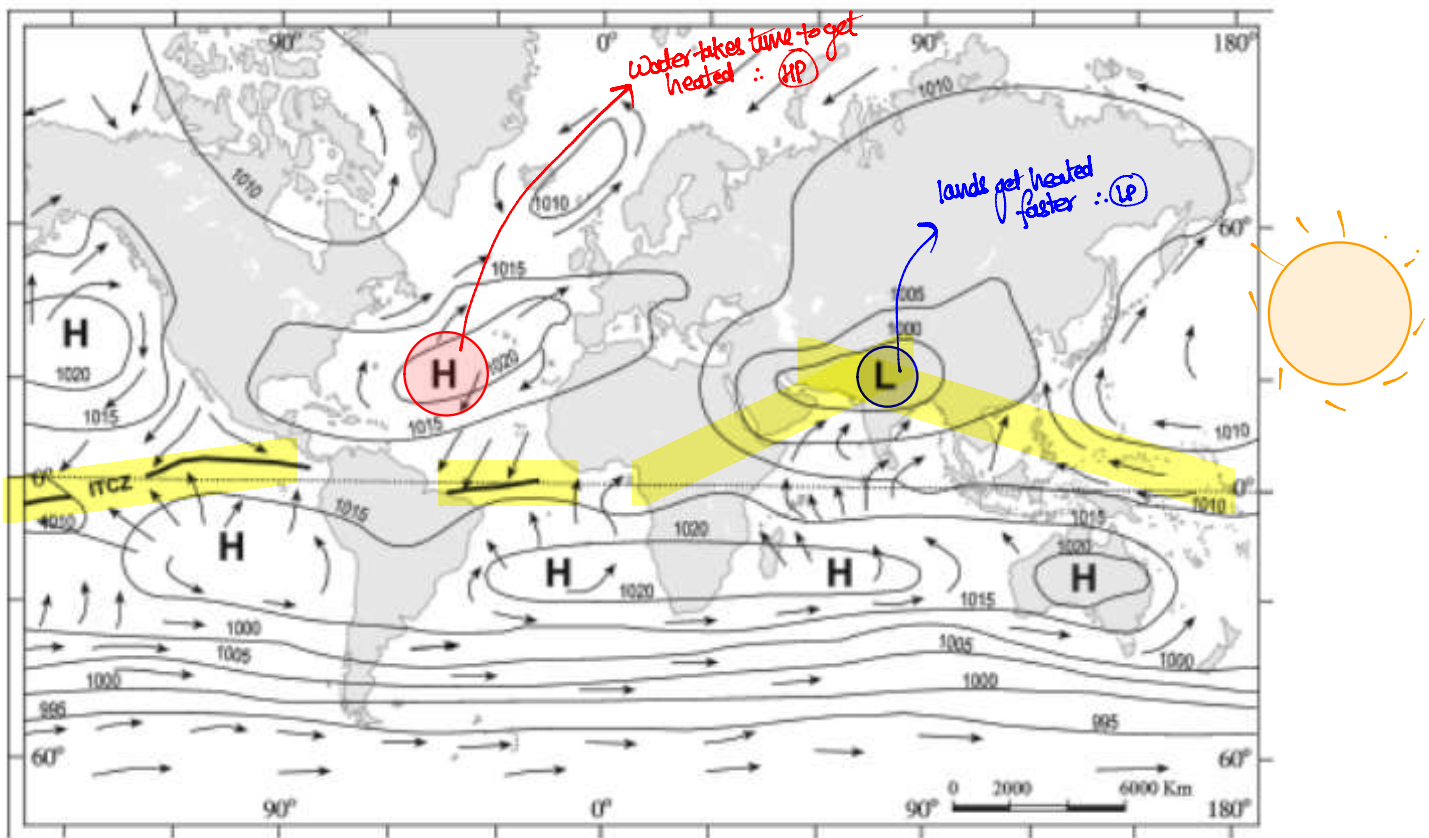


Figure 10.3 : Distribution of pressure (in millibars) — July

Summers @ N. Hemisphere

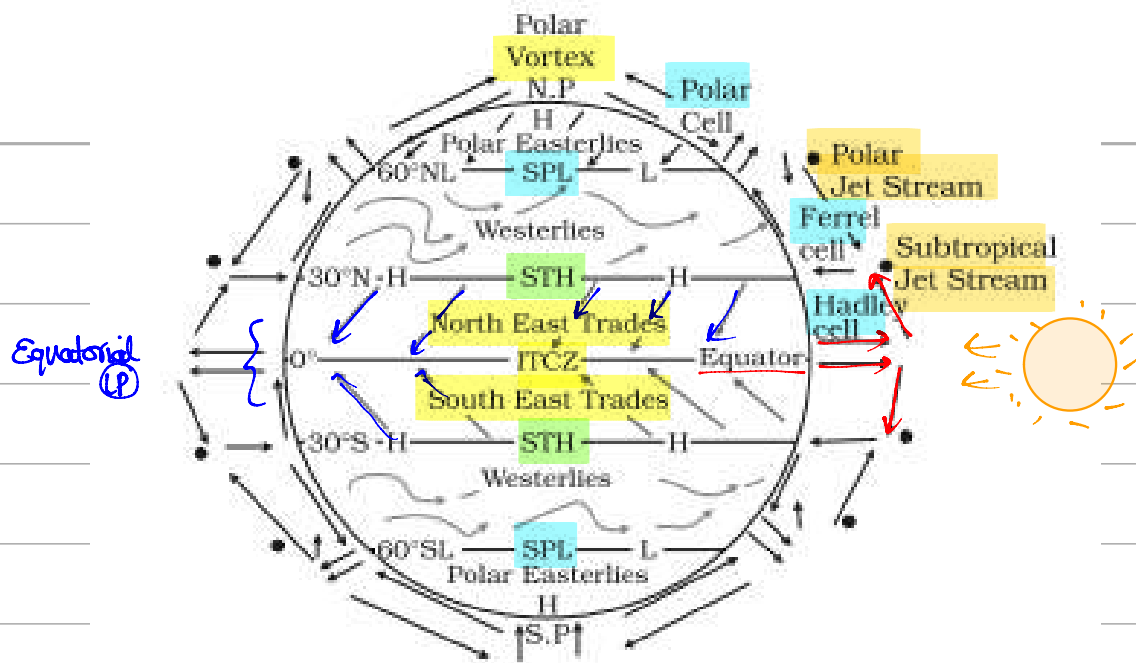
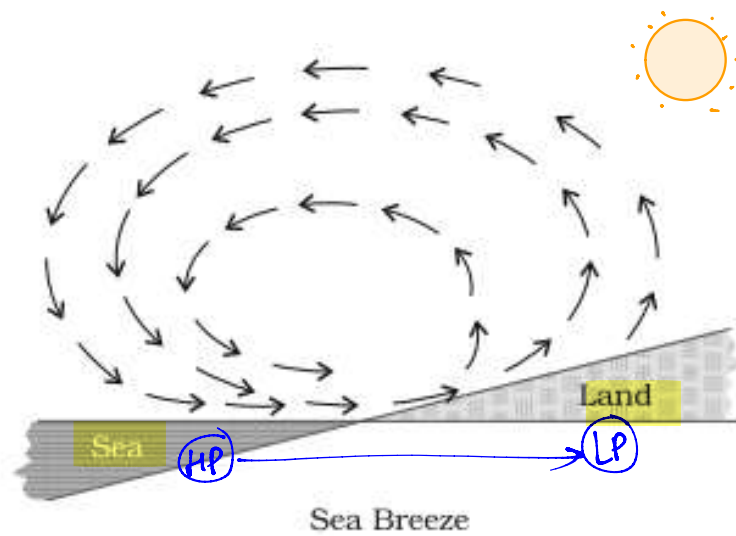
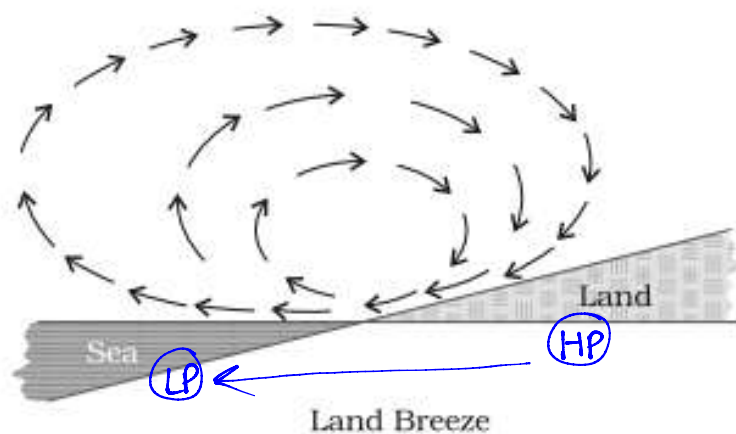


Figure 10. 6 : Simplified general circulation of the atmosphere



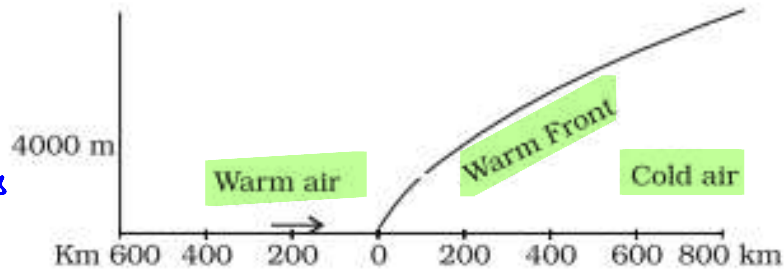
Sea Breeze



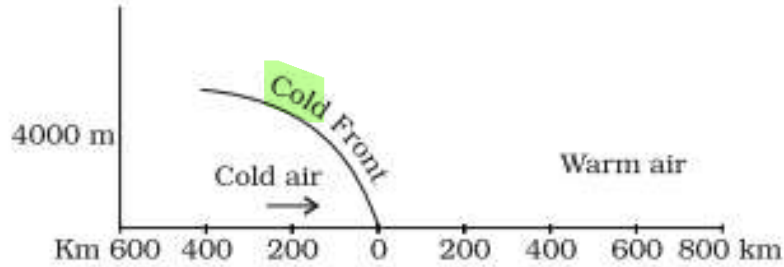
Land Breeze

Figure 10.7 : Land and sea breezes

WARM front: warm air moves towards cold air mass



Cold front: cold air moves towards warm air mass



Occluded front: air mass fully lifted above land surface.

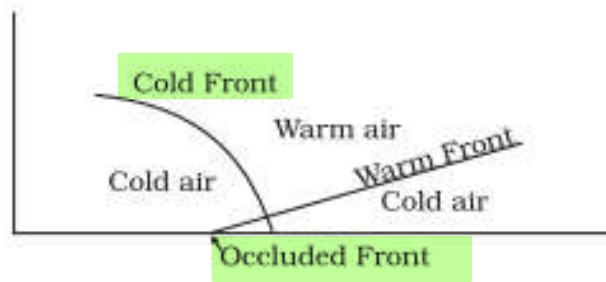


Figure 10.8 : Vertical Sections of : (a) Warm Front; (b) Cold Front; (c) Occluded Front

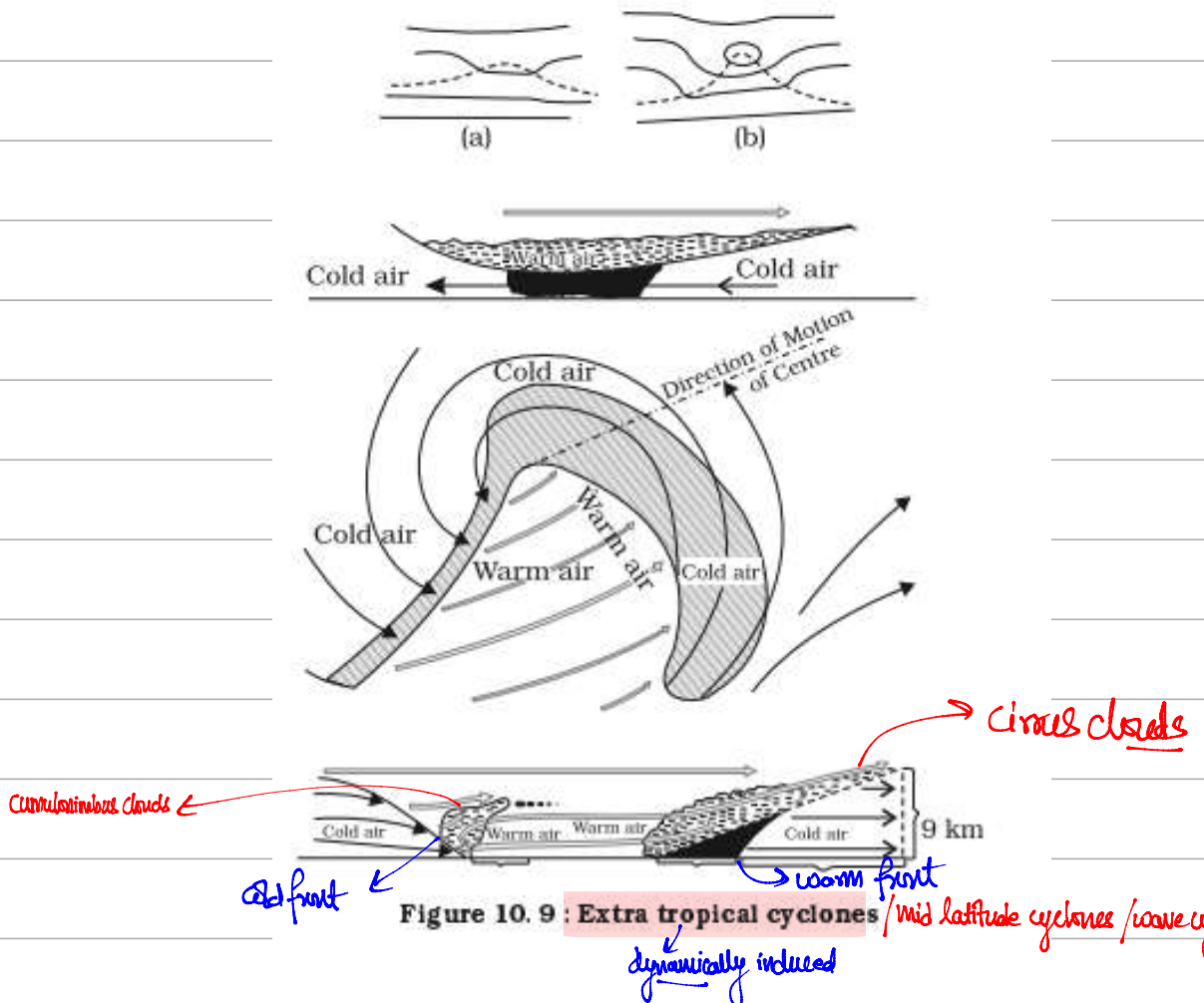


Figure 10.9 : Extra tropical cyclones

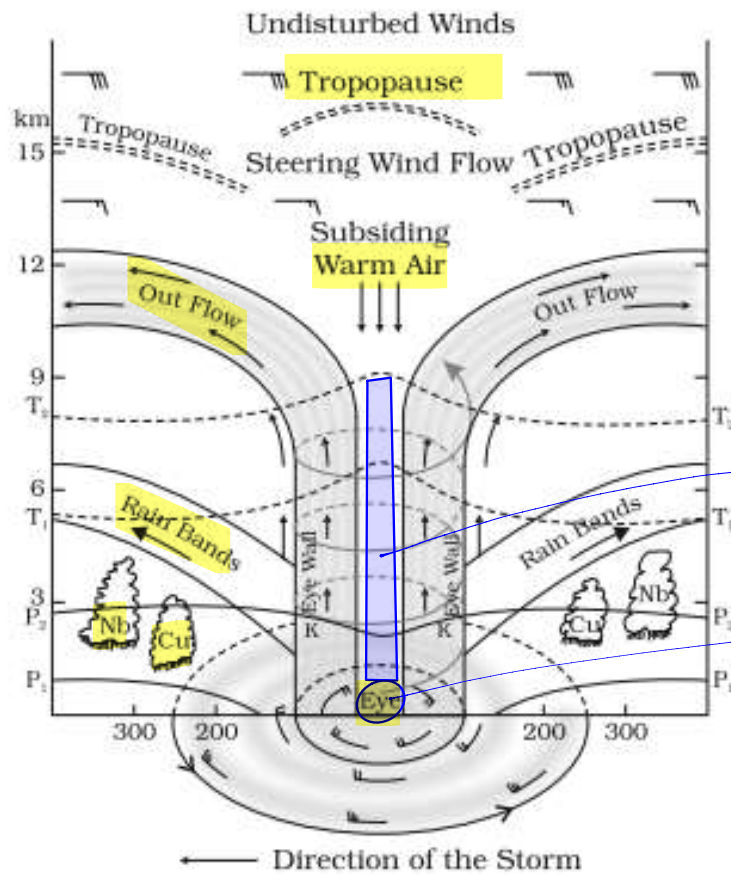


Figure 10.10 : Vertical section of the tropical cyclone (after Rama Sastry)

Table 13.1 : Water on the Earth's surface

Reservoir	Volume (Million Cubic km)	Percentage of the Total
Oceans	1,370	97.25
Ice Caps and Glaciers	29	2.05
Groundwater	9.5	0.68
Lakes	0.125	0.01
Soil Moisture	0.065	0.005
Atmosphere	0.013	0.001
Streams and Rivers	0.0017	0.0001
Biosphere	0.0006	0.00004

Table 13.2 : Components and Processes of the Water Cycle

Components	Processes
Water storage in oceans	Evaporation Evapotranspiration Sublimation
Water in the atmosphere	Condensation Precipitation
Water storage in ice and snow	Snowmelt runoff to streams
Surface runoff	Stream flow fresh water storage infiltration
Groundwater storage	Groundwater discharge springs

Table 12.1 : Climatic Groups According to Koeppen

→ Close Relationship b/w distribution of "Veget" and climate

Group	Characteristics
A - Tropical	Average temperature of the coldest month is 18° C or higher
B - Dry Climates	Potential evaporation exceeds precipitation (evapord > ppt)
C - Warm Temperate	The average temperature of the coldest month of the (Mid-latitude) climates years is higher than minus 3° C but below 18° C
D - Cold Snow Forest Climates	The average temperature of the coldest month is minus 3° C or below
E - Cold Climates	Average temperature for all months is below 10° C
H - High Land	Cold due to elevation

Table 12.2 : Climatic Types According to Koeppen

Group	Type	Letter Code	Characteristics
A-Tropical Humid Climate	Tropical wet	Af	No dry season ; afternoon ppt, Evergreen forest @ Amazon Basin, Indonesia;
	Tropical monsoon	Am	Monsoonal, short dry season ; Indian subcontinent, NE S.Am, N.Aust
	Tropical wet and dry	Aw	Winter dry season ; deciduous trees
B-Dry Climate	Subtropical steppe	BSh	Low-latitude semi arid or dry
	Subtropical desert	BWh	Low-latitude arid or dry
	Mid-latitude steppe	BSk	Mid-latitude semi arid or dry
	Mid-latitude desert	BWk	Mid-latitude arid or dry
C-Warm temperate (Mid-latitude) Climates	Humid subtropical	Cfa	No dry season, warm summer ; northern Indian plains
	Mediterranean	Cs	Dry hot summer
	Marine west coast	Cfb	No dry season, warm and cool summer
D-Cold Snow-forest Climates	Humid continental	Df	No dry season, severe winter
	Subarctic	Dw	Winter dry and very severe
E-Cold Climates	Tundra	ET	No true summer
	Polar ice cap	EF	Perennial ice
H-Highland	Highland	H	Highland with snow cover

Wet in humid
Eto Tropics
ITCZ
low annual range
low rainfall
15-65 mts
areas of sub trop high

h → wet
k → cold
S → semi Arid
W → Arid

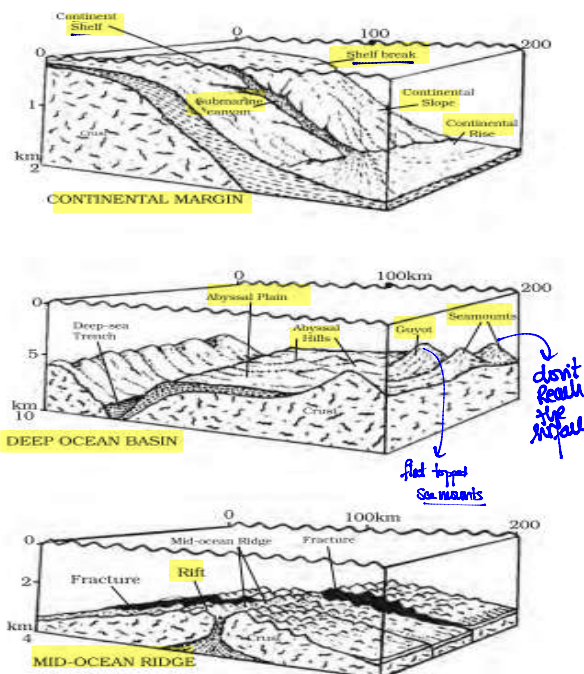


Figure 13.2 : Relief features of ocean floors

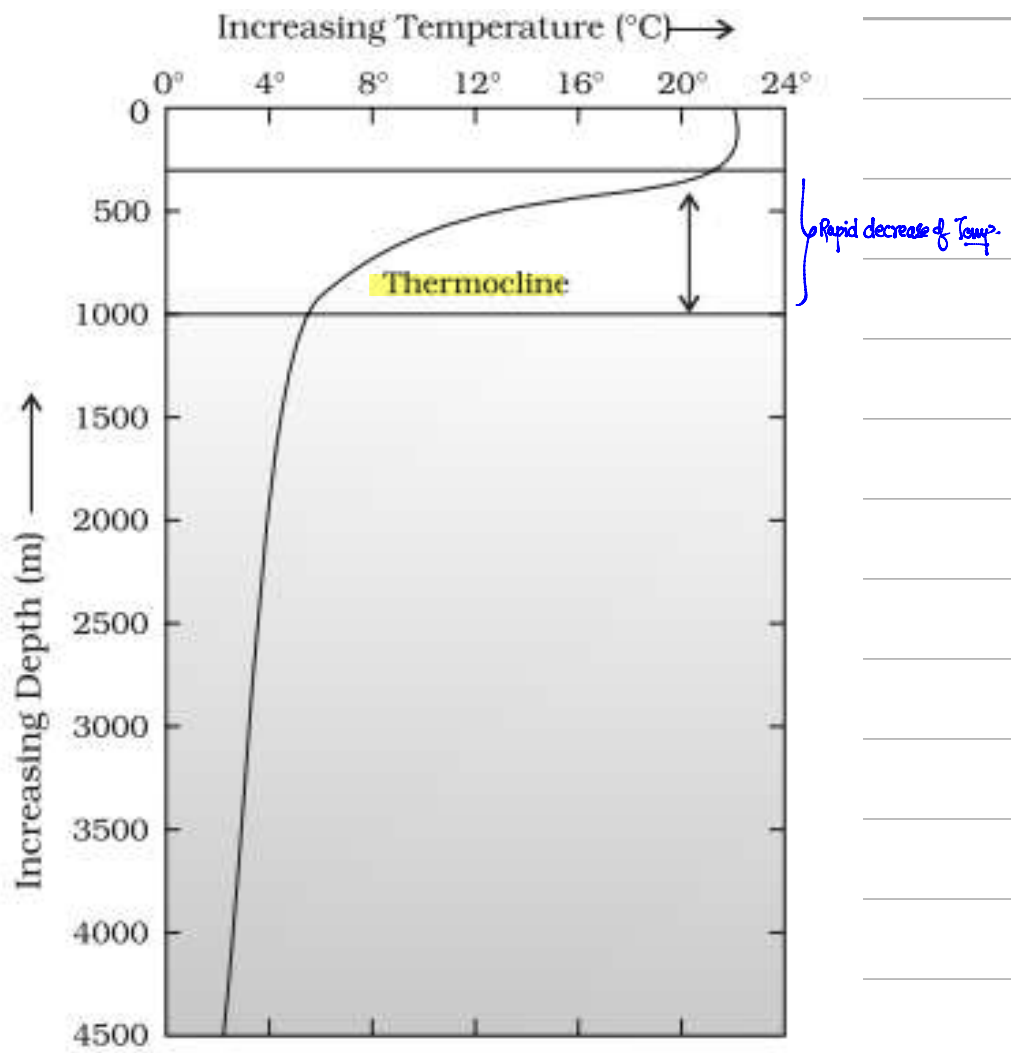


Figure 13.3 : Thermocline

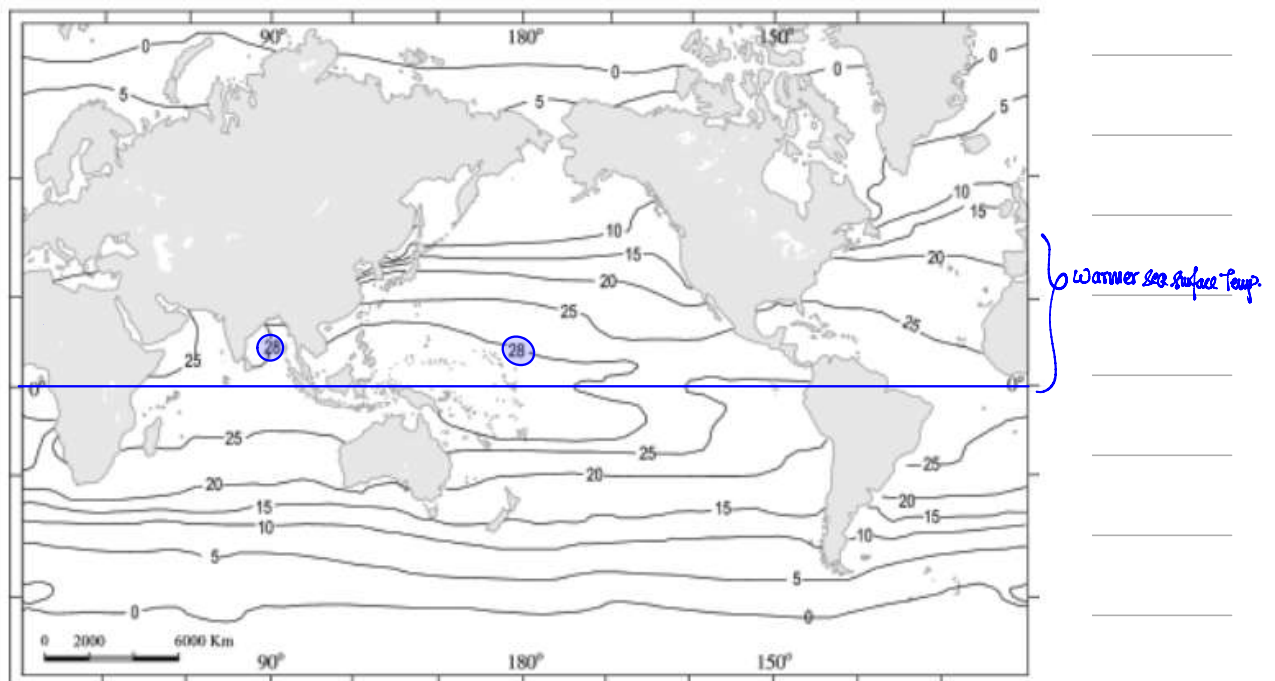


Figure 13.4 : Spatial pattern of surface temperature ($^{\circ}\text{C}$) of the oceans

Highest salinity in water bodies
 Lake Van in Turkey ($330^{\circ}/_{00}$),
 Dead Sea ($238^{\circ}/_{00}$),
 Great Salt Lake ($220^{\circ}/_{00}$)

Table 13.4 : Dissolved Salts in Sea Water
 (gm of Salt per kg of Water)

① Chlorine	18.97
② Sodium	10.47
③ Sulphate	2.65
④ Magnesium	1.28
⑤ Calcium	0.41
Potassium	0.38
Bicarbonate	0.14
Bromine	0.06
Borate	0.02
Strontium	0.01

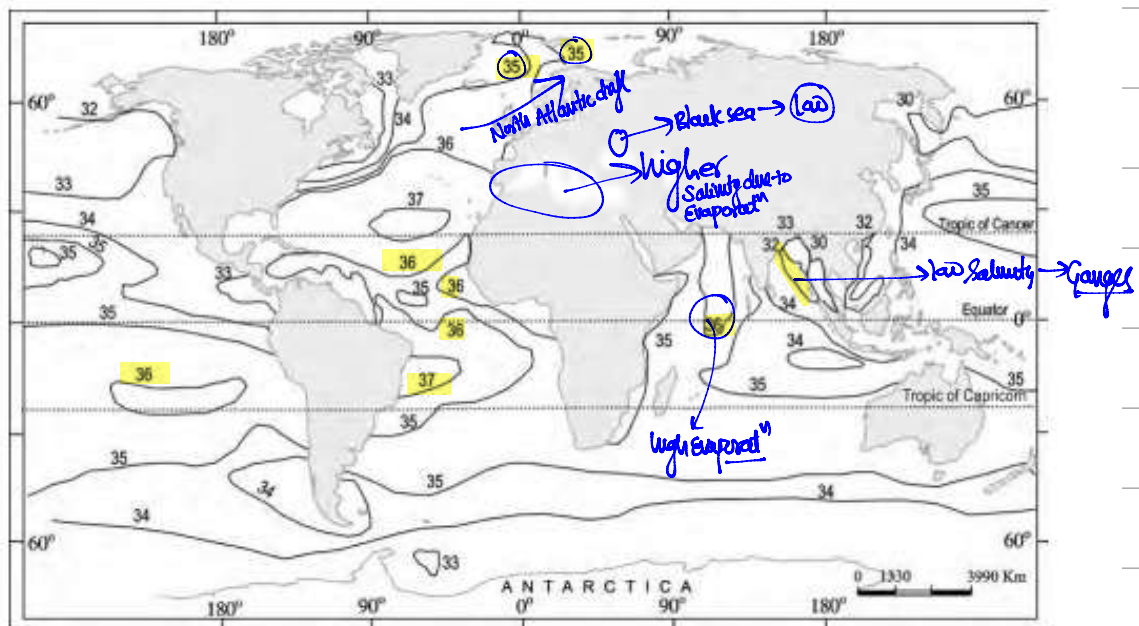


Figure 13.5 : Surface salinity of the World's Oceans

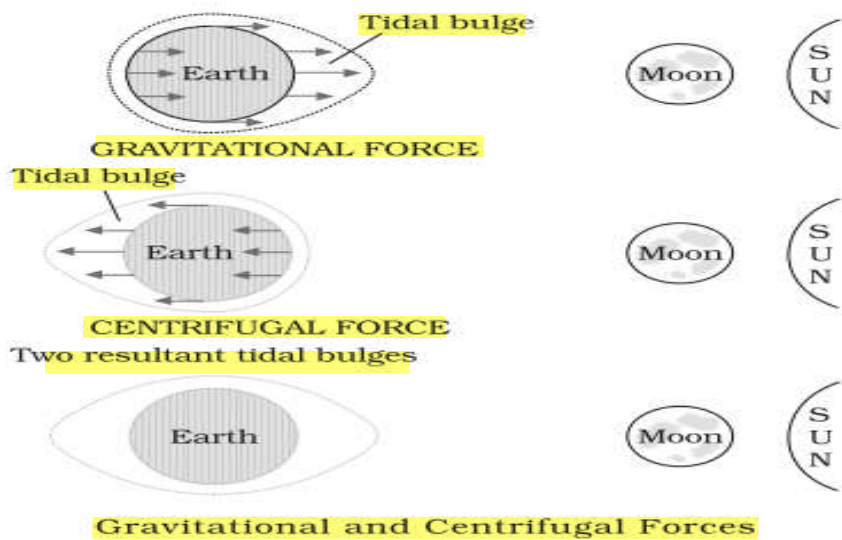


Figure 14.2 : Relation between gravitational forces and tides

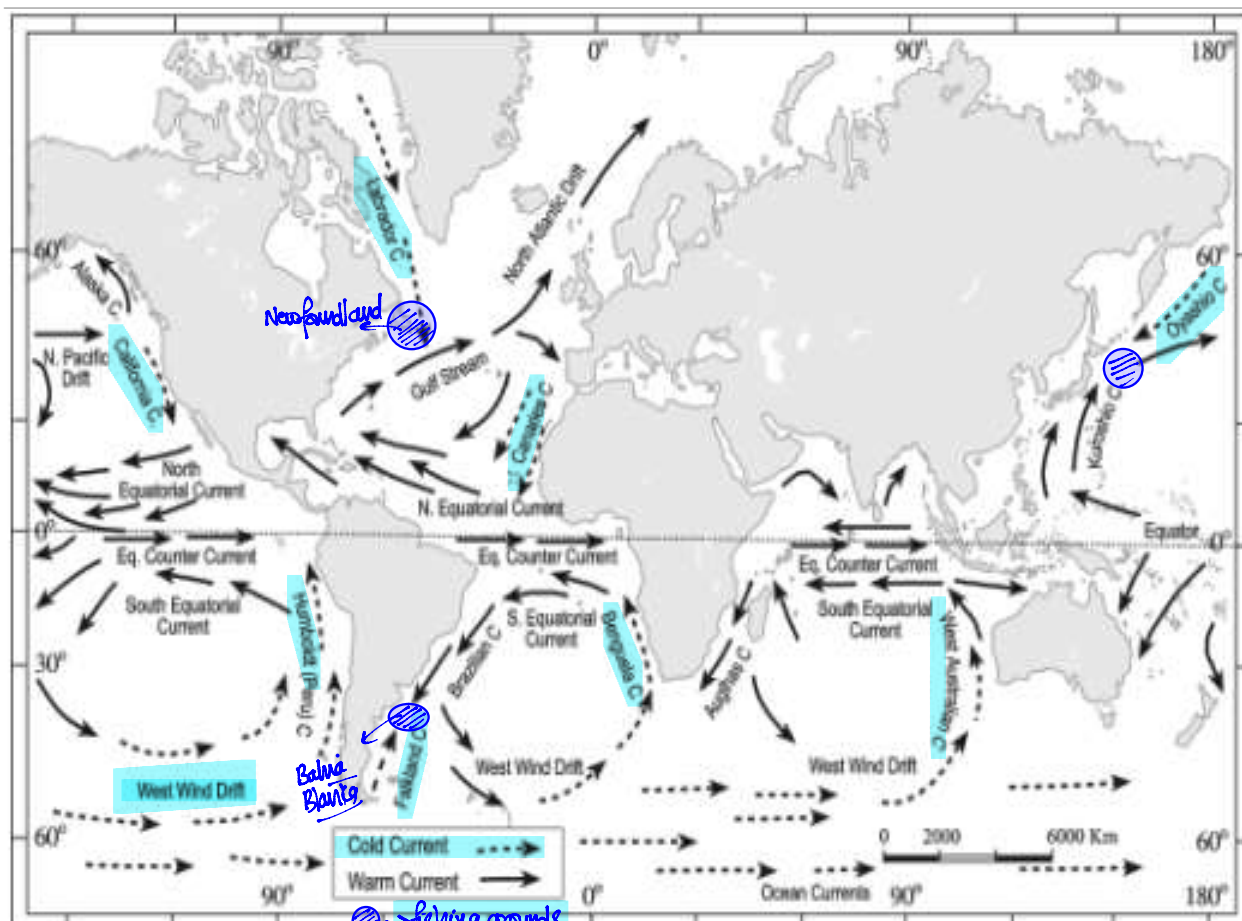


Table 15.1 : World Biomes

Biomes	Subtypes	Regions	Climatic Characteristics	Soil	Flora and Fauna
Forest	A. Tropical 1. Equatorial 2. Deciduous B. Temperate C. Boreal	A1. 10° N-S A2. 10° - 25° N-S B. Eastern North America, N.E. Asia, Western and Central Europe C. Broad belt of Eurasia and North America, parts of Siberia, Alaska, Canada and Scandinavia	A1. Temp. 20-25°C. evenly distributed A2. Temp. 25-30°C. Rainfall, ave. ann. 1,000mm. seasonal B. Temp. 20-30° C. Rainfall evenly distributed 750-1,500mm. Well-defined seasons and distinct winter. C. Short moist moderately warm summers and long cold dry winter; very low temperatures. Precipitation mostly snowfall 400 -1,000mm	A1. Acidic, poor in nutrients A2. Rich in nutrients B. Fertile, enriched with decaying litter C. Acidic and poor in nutrients, thin soil cover	A1. Multi-layered canopy tall and large trees A2. Less dense, trees of medium height; many varieties co-exist. Insects, bats, birds and mammals are common species in both B. Moderately dense broad leaved trees. With less diversity of plant species. Oak, Beach, Maple etc. are some common species. Squirrels, rabbits, skunks, birds, black bears, mountain lions etc. C. Evergreen conifers like pine, fir and spruce etc. Wood peckers, hawks, bears, wolves, deer, hares and bats are common animals
Desert	A. Hot and Dry desert B. Semi arid desert C. Coastal desert D. Cold desert	A. Sahara, Kalahari, Marushtali, Rub-el-Khali B. Marginal areas of hot deserts C. Atacama D. Tundra climatic regions	A. Temp. 20 - 45°C. B. 21 - 38°C. C. 15 - 35°C. D. 2 - 25°C A-D Rainfall is less than 50 mm	Rich in nutrients with little or no organic matter	A-C. Scanty vegetation; few large mammals, insects, reptiles and birds D. Rabbits, rats, antelopes and ground squirrels
Grassland	A. Tropical Savannah B. Temperate Steppe	A. Large areas of Africa, Australia, South America and India B. Parts of Eurasia and North America	A. Warm hot climates. Rainfall 500-1,250 mm B. Hot summers and cold winter. Rainfall 500 - 900 mm	A. Porous with thin layer of humus. B. Thin flocculated soil, rich in bases	A. Grasses; trees and large shrubs absent; giraffes, zebras, buffalos, leopards, hyenas, elephants, mice, moles, snakes and worms etc., are common animals B. Grasses; occasional trees such as cottonwoods, oaks and willows; gazelles.
					oceros, wild horses, lions, varieties of birds, worms, snakes etc., are common animals
Aquatic	A. Freshwater B. Marine	A. Lakes, streams, rivers and wetlands B. Oceans, coral reefs, lagoons and estuaries	A-B Temperatures vary widely with cooler air temperatures and high humidity	A. Water, swamps and marshes B. Water, tidal swamps and marshes	Algal and other aquatic and marine plant communities with varieties of water dwelling animals
Altitudinal	—	Slopes of high mountain ranges like the Himalayas, the Andes and the Rockies	Temperature and precipitation vary depending upon latitudinal zone	Regolith over slopes	Deciduous to tundra vegetation varying according to altitude

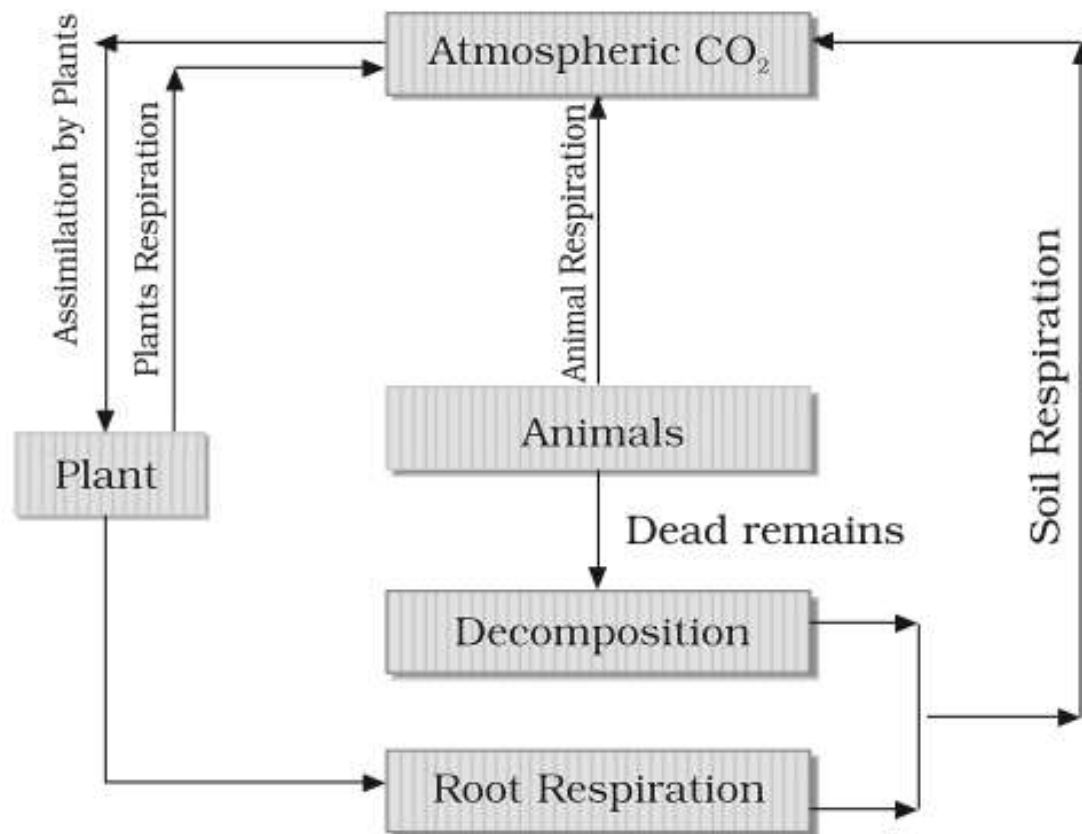


Figure 15.2 : Carbon Cycle

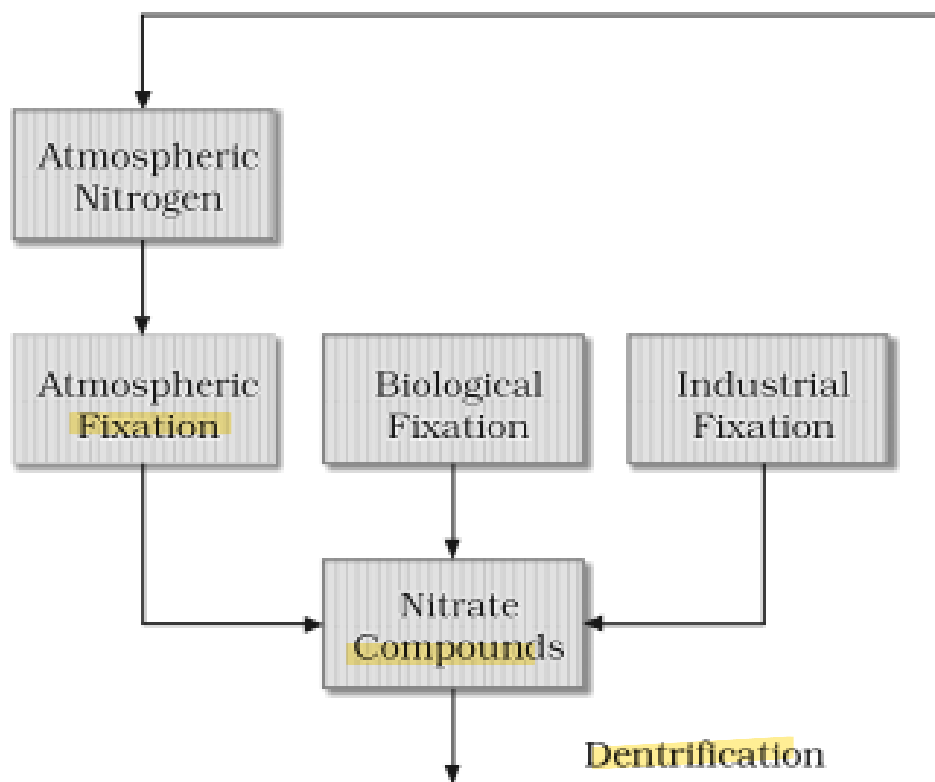


Figure 15.3 : Nitrogen Cycle

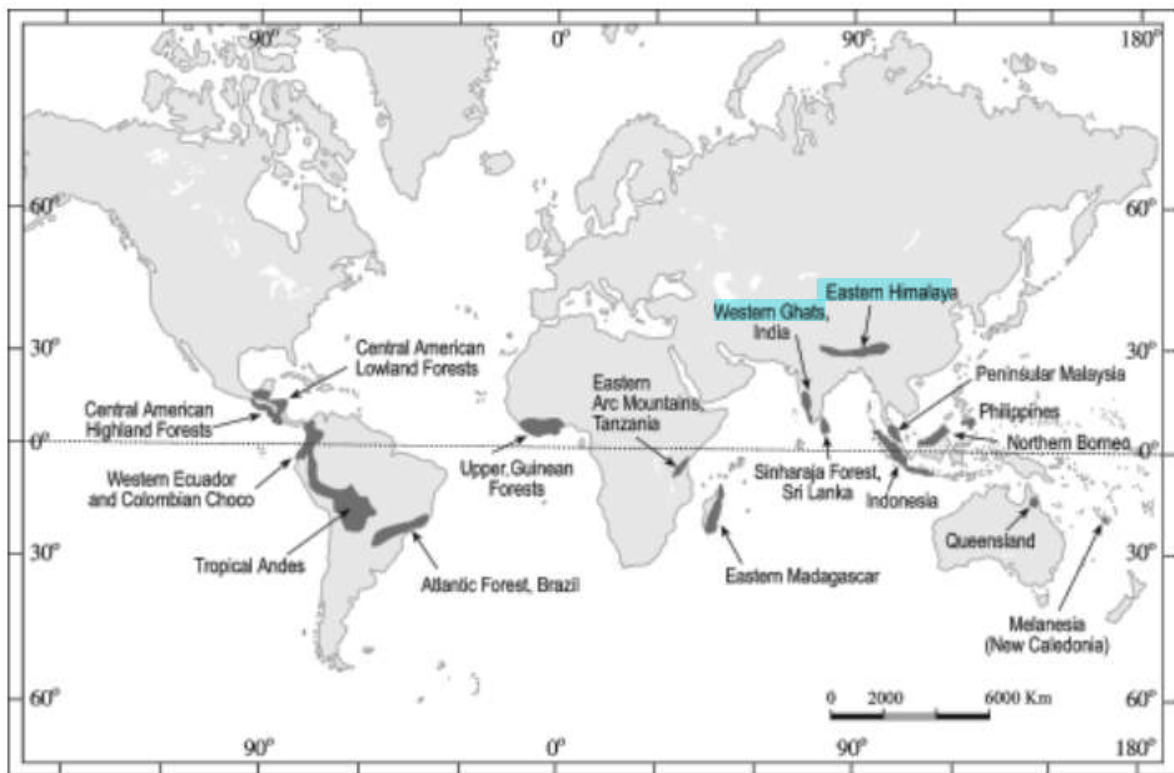


Figure 16.5 : Ecological 'hotspots' in the world